NIHR | National Institute for Health and Care Research

Reference number	NIHR207682
Programme	Policy Research Programme
Call	PRP (38-02-08) Demand for & supply of therapists for children and young people with SEND
Lead Applicant	Ms Marie Gascoigne
Host organisation	BETTER COMMUNICATION CIC
Research Title	Meeting the needs of children and young people (0-18) with Special Educational Needs and/or Disabilities: modelling the Allied Health Professional therapy workforce within integrated children's services.
Start Date	01/01/2026
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Research Type	Primary Research

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1. The Research Team

Full Name	Ms Marie Gascoigne	**Institution**	BETTER COMMUNICATION CIC

Specify your (lead applicant) role in the research
Marie Gascoigne is a speech and language therapist and Director of Better Communication CIC, a not-for-profit social enterprise. Marie has worked in clinical practice, higher education, research, management and policy development for over 30 years. Marie has authored many policy reports and been part of national reviews around the commissioning and provision of therapy services. Marie has developed the Balanced System® Outcomes Framework whole system approach for supporting therapy needs as part of the integrated workforce. Marie will oversee the use of the Balanced System for WP3 and will be the main link to the ICBs and Local Authorities nationally.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project. If you are funded as part of other NIHR projects that will be running concurrently, your time must not exceed 100% overall.
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Joint Lead Applicant

Full Name	Professor Victoria Joffe	**Position**	Professor and Dean
Department	School of Health and Social Care	**Institution**	University of Essex

Justification for Joint Lead Applicant
Professor Victoria Joffe is a a specialist speech and language therapist working with children and young people with special educational needs and disabilities. She is expert in developing evidence-based interventions for children and young people with Speech, Language and Communication Needs (SLCN) and in workforce development and training in the health and social care arena. Joffe has experience leading on and participating in large scale randomised control trials, incorporating mixed methodologies and drawing on qualitative and quantitative analyses. She is experienced in developing national and international surveys, and running focus groups and interviews. She will co lead the research programme with Gascoigne overseeing all work packages, and will lead on work package one. Joffe works closely with NHS E, integrated care boards, providers and commissioners, in supporting children and young people with SLCN, and in developing an appropriate Allied Health Professional workforce. She will draw on these partnerships to facilitate recruitment of all stakeholders regionally and nationally. She will support and mentor the research associates on the project, and input into the qualitative and quantitative data collection and analyses. She will co lead with Gascoigne on overall management of the project, plan with the PPIE lead the engagement activities, and lead on overall dissemination of the research. With Gascoigne, she will oversee the development and running of the project steering group, manage the project team and operational management and governance of the research programme and liaise closely with the Department of Health and Social Care and the policy team.

Relevant expertise and experience of Joint Lead Applicant
Professor Victoria Joffe is a highly specialist speech and language therapist, and professor of Speech and Language Therapy, with over 30 years experience in working with children, young people and adults with special educational needs and disabilities (SEND). She is Dean of Integrated Health and Care Partnerships at the University of Essex, having developed strong partnerships with Integrated Care Boards, NHS E, local authorities, providers, health and education commissioners and third sector organisations, as well as with parents and families of children with SEND. She has extensive experience working in speech and language therapy and the wider allied health professional arena. Joffe has expertise in workforce planning and is involved in the training of allied health professionals, and in the evaluation and modelling of best practice across service delivery models for children and young people with

SEND. She will draw on her extensive partnerships to facilitate recruitment of all participant groups and ensure involvement and engagement of key stakeholders. Joffe has had experience in leading a large randomised control trial with young people with speech, language and communication needs, and has been a co-investigator of several NIHR-funded feasibility and large intervention studies with children with social communication difficulty, Down Syndrome and with young people who stutter. She is experienced in conducting and completing large mixed method research studies on time, developing qualitative and quantitative research methodologies and appropriate analyses, and ensuring engagement of and dissemination to all stakeholders, with appropriate outputs for varied audiences: scientific, policy, professional and general public.

Specify role in research
Joffe, together with Gascoigne, will lead the co-ordination and implementation of the research, with oversight of research governance, organisation and management. She will oversee all work package 1, and ensure project deliverables are met within the timeframe. She will co-ordinate the project steering group, and project team, ensuring engagement of stakeholders at all stages of the research. She will oversee the academic, policy and societal impact of the research, and communication and dissemination of results.

%FTE commitment	10

Co-applicant role and % FTE commitment
Specify the role and percent FTE commitment of each co-applicant.

Full Name	Dr Emma Pagnamenta	**Institution**	University of Reading

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Emma Pagnamenta, with vast experience working with families of children with SEND, is PPIE lead. Emma has worked extensively with the Public Co-applicant for this research to shape the research plan and will continue to support them in that role as well as leading engagement with the group of children and young people and families who will form the PPIE reference group. Emma will also support dissemination that is co-designed with children and young people.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.
10

Full Name	Professor Reinhold Scherer	**Institution**	University of Essex

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Scherer will lead the project's statistical and quantitative modeling, applying advanced techniques in artificial intelligence, machine learning, and healthcare simulation. He will oversee the design and execution of data analytics strategies, ensuring methodological rigor and actionable insights. His role includes integrating complex datasets,

developing predictive models, and supporting evidence-based decision-making. He will work with the Lead Investigator, Gascoigne, to develop the workforce modelling within the Balanced System.

% FTE
Commitment: This refers to the percentage of your time that you will commit to this project.

10

Full Name	Mrs Sharon Tuppeny	**Institution**	BETTER COMMUNICATION CIC

Please indicate if this co-applicant is a member of the public

No

Specify role in research

Sharon Tuppeny is a highly specialist children's occupational therapist who has extensive experience in working with children and young people through the lens of a whole systems approach. Sharon leads the support for occupational therapy teams redesigning their service provision towards a Balanced System approach. Sharon will be closely involved across all work packages and as well being closely involved in WP3 leading on data collection from an occupational therapy perspective.

% FTE
Commitment: This refers to the percentage of your time that you will commit to this project.

8

Full Name	Miss Maxine Darby	**Institution**	BETTER COMMUNICATION CIC

Please indicate if this co-applicant is a member of the public

No

Specify role in research

Maxine Darby is a highly specialist children's physiotherapist who has extensive experience in working with children and young people through the lens of a whole systems approach. Maxine leads the support for physiotherapy teams redesigning their service provision towards a Balanced System approach. Maxine will oversee the review and collection of data required for this element of the modelling in WP3 as well as contributing physiotherapy expertise across the whole research project.

% FTE
Commitment: This refers to the percentage of your time that you will commit to this project.

5

Full Name	Dr Hortensia Gimeno	**Institution**	Queen Mary University of London

Please indicate if this co-applicant is a member of the public
No

Specify role in research
Dr Hortensia Gimeno, clinical academic paediatric occupational therapist and NIHR Advanced Fellow, Clinical Reader in Child Health Research, Associate Director for Therapies Research. With 25+ year's experience with children with complex conditions. She has expertise in community based children's therapy services, mehtodology expertise in scoping reviews, mixed-methods, qualitative and creative methods with children, PPI and ethics. Gimeno will lead workpackage 2 and contribute to results interpretation for all work packages, and dissemination.

% FTE Commitment: This refers to the percentage of your time that you will commit to this project.
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Full Name	Miss Sophia TA Hill	**Institution**	PPI representative

Please indicate if this co-applicant is a member of the public
Yes

Please include a clear description of their role and the reasons why a public co-applicant is joining the team. Co-applicants who are patients, service users or carers are not obliged to complete a standard CV but are required to provide a summary of any knowledge, skills and experience relevant to their role in the application.
Sophia brings lived experience of SEND and multiple therapies to the research team. Feedback at Stage 1 highlighted that the voices of children and young people did not feature strongly enough. Sophia's role as a co-researcher with lived experience, and as co-PPI lead, alongside EP, will ensure that young people are at the centre and that accessible and inclusive research practices are used throughout. Sophia has an impressive array of experiences of working with stakeholders to share her lived experience and bring about change. Sophia has supported teachers and teaching assistants to gain an awareness of lived experience of SEND as part of staff induction processes, including the delivery of presentations. Sophia is also a member of a council-wide SEND young people's group, called Impacts. This has enabled involvement in and feedback to Devon-wide activities including Education Health and Care Plan technology and information, educational psychology information about neurodiversity and neurodivergence and supporting strategies in schools and a SEND young people survey. Sophia has extensive knowledge and experience of physiotherapy, occupational therapy and speech and language therapy. This includes experience of ineffective referral pathways and waiting lists in addition to effective service provision that has brought about positive outcomes in mobility, confidence, reduced pain and discomfort, engagement with daily living activities and self-advocacy. Sophia has a key understanding around barriers to inclusion for young people with SEND which will be invaluable in using creative and support strategies to work with children and young people in our PPI group and also in delivery of WP2. Sophia will support recruitment of other public contributors (young people and parents), work with EP to coordinate and deliver PPI activities (e.g. PPI group meetings, training of PPI groups), and support data collection and analysis, particularly in the design and delivery of workshops with children and young people and families (WP2). Sophia will also contribute to training and recruiting new researchers based on lived experience, and will have a key role in our dissemination strategy. This will include supporting the co-production of a public-facing film/podcast and easy read summaries of results, co-presenting research results at dissemination events and contributing to articles and research summaries.

Specify role in research

As public co-applicant Sophia will oversee the conduct and progress of the study, contribute directly to all stages based on lived experience and co-design and deliver some aspects of the study including WP2, and as co-lead on PPI, PPI groups and public involvement activities. Sophia will have a key role in dissemination and outputs.

% FTE
Commitment: This refers to the percentage of your time that you will commit to this project.

5

2. History of application

Has this application been previously submitted to this or any other funding body?

No

3. Scientific abstract

Scientific abstract

An estimated 1.7 million children and young people (CYP) in England are identified as having special educational needs and disabilities (SEND). The 2024/25 Education Select Committee inquiry, “Solving the SEND Crisis”, is examining the barriers to achieving a whole system approach; The Kessler Review (2024) of the demand and supply of therapists for CYP with SEND also identified need for systemic approaches such as the Balanced System®.

Aims and Objectives

The **MO**delling **T**herapies for **I**mproved **F**utures (MOTIF) study aims to inform policy, practice, workforce planning and commissioning for therapy services supporting CYP with SEND (0-18) based on a needs-led model.

Objectives to:

1. Examine key risk factors that contribute to the needs of CYP with SEND for therapy services.
2. Review the policy and strategic guidance relevant to the funding, demand and supply of the relevant therapy workforce
3. Gather views regarding current therapy practices and the acceptability of a systems-based delivery model from (i) therapy workforce (ii) professional associations and (iii) CYP and Families
4. Identify the enablers and challenges for commissioning of therapies for CYP with SEND in two selected regions: Greater Essex and Midlands
5. Generate therapies workforce models based on population need building on existing data and methods from a systems model (the Balanced System®) enhanced and further developed through this research programme.

Methods

MOTIF will be delivered in three work packages (WPs):

WP-1: Evidence synthesis in the form of two scoping reviews i) risk factors determining population need for support in CYP with SEND, and ii) analysing policy documents to understand contextual factors impacting on workforce modelling and delivery of therapies (Objectives 1-2)

WP-2: Multi-Stakeholder Engagement to gather qualitative and quantitative data through focus groups, surveys with professionals, and creative methods participatory workshops with CYP, parents/carers to identify perspectives on what is working well within current provision models and acceptability of a systems-based model (Objectives 3-4)

WP3: Systems-thinking approach to workforce modelling (Objective 5)

Analysis

Quantitative data will be analysed using descriptive statistics. Qualitative data will undergo thematic analysis (focus groups) and content analysis (workshops). Workforce modelling will be iterative throughout the course of the programme, starting with refinement and development of existing modelling of population need for speech, language and communication support to include the broader group of CYP, enabling calibration of the applied workforce model to a range of populations within the two target regions and acceptability testing with stakeholders.

Anticipated outputs:

A population needs profile for therapy support for England

A workforce model for therapy provision tested in two regions and ready for future testing and validation across England.

Dissemination through peer-reviewed publications, lay summaries, public facing video, podcasts / media developed with public contributors, accessible materials

PPI/EDI:

Our public co-applicant (a young person with SEND and experience of multiple therapies) and 5 parents of CYP with SEND have supported the development of this application. PPI groups will include parents/carers and CYP with SEND and will work closely with the research team to inform the design and delivery of all work packages.

4. Plain English Summary

Plain English summary

What is the problem?

There are many children and young people who need support to learn at school and to achieve their potential. Speech and language therapists, occupational therapists and physiotherapists are part of the team that can help. The way support happens now isn't working for many children and young people who have special educational needs and disabilities. The way we deliver therapy must change.

Why is this topic important?

The government wants to understand how many therapists are needed to provide the right support and therapy for children and young people. We want to understand if we need more therapists in some areas of the country.

What is the research aim?

Our aim is to help plan therapy that supports children and young people with special educational needs and disabilities. We want to make sure they get support that will make a difference. This will help us decide what support should be paid for and given.

What is the research plan?

We will:

1. Look at reports from the government and other organisations that pay for special educational needs services and published research. This will help us to understand who needs what type of support and if services are helping children and young people in the ways they need.
2. Send surveys to health professionals asking them what they think currently works well and what things aren't happening right now that should be happening. We will also talk to people who decide how much money should be spent on what services.
3. Talk to children and young people as well as their families to learn what works for them. We will ask them what they think of working in different ways – for example, working in a group with other children, or getting help in the class at school, or changing things that will help a lot of children such as training school staff.

Who will do the research?

Our team includes health professionals and researchers who are experts in occupational therapy, physiotherapy and speech and language therapy for children and young people with special educational needs and disabilities.

How will children and parents be involved?

A young person with lived experience of special educational needs and therapies is on our team and has helped to develop this application. We will have a group of parents and children and young people who will help us to design and deliver the project from start to finish.

How will we share the results?

We will work with children and young people and parents to create easy-to-understand ways to share the findings, like a video or podcast. We will also share results through scientific papers, charities, or other channels, for example, parent/carer forums.

5. Changes from first stage

Changes from Stage 1

Changes since Stage 1 application have been significant. The team felt unable to submit the Stage 2 application in May 2024 for a number of reasons. First, the the scope of the grant call was very broad and it was difficult to ascertain what the main focus would be. Second, the timeframe and overall budget appeared to the research team to be prohibitive and implausible with regards to responding adequately to the call and producing a high-quality proposal.

In the intervening 12 months, the team have been in discussion with the NIHR and DHSC client in order to shape a proposal that both achieves the ultimate outcome of providing a workforce modelling approach for all three therapies (speech and language therapy, occupational therapy and physiotherapy) for children and young people with special educational needs and disabilities and is feasible within the overall budget and a two year time frame for delivery.

Discussions have focused on the essential outcomes that the Policy Research Programme wish to facilitate for the policy teams within DHSC and DfE, namely a robust methodology for workforce planning for children's speech and language therapy, occupational therapy and physiotherapy. The content of the research plan has been honed towards ensuring that all the activity is very directly focused on essential research to support modelling.

In the wider context, the intervening year has seen change in Government and a number of enquiries into the SEND system in which the supply of therapy support has featured significantly. Reports of the Public Accounts Committee and National Audit Office have both fed into the ongoing Education Committee enquiry, Solving the SEND Crisis.

Marie Gascoigne was asked to give oral evidence to the Education Select Committee and asked specifically about the potential impact of the Balanced System as a whole system approach. Research from the University of Warwick for 'What works in SEND' highlighted the value of the Balanced System as one of five approaches identified in a review. The Committee comments regarding the Stage 1 application questioned whether the Balanced System outputs would be generalisable to occupational therapy and physiotherapy. The Warwick review includes examples of impact data from areas where the Balanced System is being used currently to transform all three therapy systems and also an example of it's use as the underpinning framework for the SEND Inclusion Framework in a Local Authority. These outputs are emerging from projects that have matured in the intervening year since May, 2024 and provide evidence of transferability.

This Stage 2 application is therefore in the context of the feedback from the Committee to the Stage 1 application, the iterative refinement of the brief over the past year, and the changing external policy context for this particular research.

Two outline proposals were considered by the NIHR / DHSC team one in July 2024 and a further refinement in November 2024. This Stage 2 application is based on the preferred option from three that were proposed in November, 2024. The key elements of the chosen option were,

1. Ensuring that the outcome was a modelling methodology that would enable occupational, participatory and physical needs of children and young people to be considered equally alongside speech, language and communication needs when modelling workforce.
2. Retaining strong stakeholder engagement across the systems from national bodies, through to local strategic systems and operational delivery with children and young people and their families central to this research
3. Focusing on depth of modelling in two areas with the compromise of geographical spread and broader validity checking within this research programme but producing a modelling outcome for all three therapy areas that has the potential to be applied nationally

As part of these iterative discussions, it was decided that Better Communication CIC would take the role of lead organisation with Marie Gascoigne as Lead Primary Investigator with Professor Victoria Joffe as Co-Lead PI. Other changes to the research team include the inclusion of Dr Hortensia Gimeno, an occupational therapist and NIHR Fellow based at Queen Mary University of London and Ms Maxine Darby, a highly specialist physiotherapist who both leads an NHS team and provides consultancy expertise via Better Communication CIC for services using the Balanced System to transform physiotherapy services.

Professor Scherer and Dr Pagnamenta continue to be co-applicants. Dr Lyons, Ms Gridley, Ms Carling are no

longer involved as co-applicants due to a re-evaluation of the expertise required to undertake the project..

The time commitments have all been adjusted to reflect the Committee's feedback. The Research Assistant time has been re-apportioned to take account of the greater involvement of Co-Apps Sharon Tuppeny and Maxine Darby in supporting the data collection using the Balanced System and in order to maximise the budget.

The Committee commended the inclusion of Public and Patient Involvement and Engagement (PPIE) in the Stage 1 application but also provided valuable feedback which we believe have been addressed in this Stage 2 application.

Engagement of PPI members throughout is described in full under 'Patient and Public Involvement' and throughout the research plan. Our PPI work in the development of this application has informed all of the work packages and outputs, methods of recruitment and involvement. Detail on numbers and recruitment is included in the research plan.

Our public co-applicant (SH), who is young person with SEND and lived experience of multiple therapies has now joined the research team. Eight young people with experience of SEND and therapies will be involved in PPI groups in each region and work with EP, SH and the research team to design and deliver each work package. SH has contributed to the development of the application and the involvement of CYP in the work packages. CYP will be involved as participants in workshops in WP2 to find out what they think would make the workforce better able to support CYP with SEND and their families. Dr Emma Pagnamenta, our PPIE lead and specialist speech and language therapist, will work with local SEND group leaders to enable participation of CYP in PPI groups.

The main changes to this Stage 2 research plan relative to the Stage 1 application are summarised as follows and are naturally outlined in detail in research plan.

We have designed an iterative approach which allows the stakeholder engagement to both inform and be informed by the modelling process. The Stage 2 application includes a visual graphic to illustrate the interaction between the three work packages.

1. The desk top reviews have been narrowed to addressing two specific information gaps to support modelling:
    a. The population risk factors and epidemiology and prevalence relating to the children and young person needs that may benefit from the involvement of occupational therapists and physiotherapists as well as speech and language therapists. The output of this review is essential to calibrating a predicted need at population level to then use in context in the case study areas.
    b. A review of the policy context for children and young people with SEND since 2015 which covers the period from the introduction of the current Special Educational Needs and Disability Code of Practice which has defined the SEND system that is currently the subject of scrutiny. This is a rapidly evolving context which is expected to have further evolved by the time this research may take place and therefore this is critical to any modelling of workforce as the SEN needs of children and young people are relative to their contexts.
2. The stakeholder engagement has also been fine tuned to focus on the essential stakeholder views and experiences to inform modelling. This element now removes the Delphi analysis from the research plan. The Committee rightly identified that this element of the Stage 1 proposal was a significant work package given the time and scope of this grant. This Stage 2 application has therefore reworked the stakeholder engagement to focus on two national and two regional engagement studies.
    a. A national survey of practitioners (speech and language therapists, occupational therapists and physiotherapists)
    b. A national level focus group for professional associations across the system including therapy professional bodies but also education and special educational needs associations
    c. Regional participatory workshops for children and young people and families in the two regions of focus
    d. Regional commissioning focus groups to include representation from all parts of the system that fund therapies for children and young people with SEND
3. The use of the Balanced System throughout this research has allowed for an iterative process with work package 3 running alongside both the desk top reviews and stakeholder engagement. The visual included in the Stage 2 application clarifies how this interaction develops the modelling. We have included more detail on the outputs that will be available to the Policy team and the timing of the various outputs.

The modelling work package has three distinct phases,

1.
    a. Modelling on predicted need for all therapies – informed by the desk top review
    b. Analysis of the contextual factors in two regions of England providing a range of diversity, economic, linguistic, ethnic, rurality – informed by the stakeholder engagement and providing the variables for modelling workforce to meet need
    c. Detailed workforce modelling in two regions bringing together the needs profile, local context including models service delivery, and local interpretation of national policy and guidance

6. Detailed Research plan

Detailed Research Plan

WHAT IS THE PROBLEM BEING ADDRESSED?

An estimated 1.7 million children and young people (CYP) (0-18 years) in England have special educational needs (SEN)[1].

The overall proportion of CYP identified as having SEN, whether supported through SEN Support or an Education and Health Care Plan, has risen from 14.4% in 2016 to 18.4% in 2024, particularly CYP with SEN requiring an Education, Health and Care Plan; 19.3% in 2016 compared to 26% in 2024)[1, 2]

The Special Educational Needs and Disability Code of Practice (0-25)[3] defines SEN as follows,

"A child or young person has SEN if they have a learning difficulty or disability which calls for special educational provision to be made for him or her"

Furthermore,

"has a significantly greater difficulty in learning than the majority of others of the same age, or has a disability which prevents or hinders him or her from making use of facilities of a kind generally provided for others of the same age in mainstream schools or mainstream post-16 institutions"[3]

The Code elaborates that many CYP will have a disability as defined by the 2010 Equality Act and that many of these will have SEN, but this is not a given. Whilst some conditions will automatically qualify as a 'disability', and many of those CYP will have SEN, there will be some for whom needs will be met by the 'ordinarily available provision' in schools and settings. For those requiring more support, there is no automatic relationship between a specific diagnostic category and whether a CYP requires an Education and Health Care Planas opposed to SEN Support.

Since introducing the Code of Practice, the term 'Special Educational Needs and / or Disability' (SEND) has been used interchangeably and variably with Special Educational Needs (SEN) for data collection. There is no distinct national data set differentiating numbers of CYP with *SEN and no disability* from those with *SEN and disability* or indeed those with *disability but no SEN*. ***For this research program, the term SEND indicates the broadest definition.***

Most CYP with SEND are educated in mainstream schools, with a smaller number in special schools. There are varied practices and marked inconsistencies in service provision for CYP with SEND across professionals, schools, and local authorities[4]. Unmet need and limited availability of services for CYP with SEND have been reported[5, 6] Challenges of access to therapy services are evident with the numbers of CYP reported to be waiting for speech and language therapy (SLT) in January 2025 reaching 66,612, with 17,838 and 15,953 waiting for Occupational Therapy (OT) and Physiotherapy (PT) respectively[7].

The 2024 Lord Darzi rapid review of the NHS identified 11% of the health budget spent for CYP, despite CYP representing 25% of the population. This funding is critical to ensuring a sufficient therapies workforce, which the Department of Health identified in June 2023 as needing substantial growth over the next decade. Notably, there are significant gaps in OT and SLT, with only limited workforce growth projected in these areas.[8, 9]

WHY IS THIS RESEARCH IMPORTANT?

The Education Select Committee inquiry, ***"Solving the SEND Crisis"*** is expected to report by the end of 2025 [10]. The inquiry is examining the barriers to achieving a 'whole system approach' to meeting the needs of CYP with SEND, exploring accountability across the health, care and education systems and seeking examples of effective practice. This follows recently published reports by the National Audit Office and the Public Accounts Committee[11, 12]. Both highlighted financial unsustainability of the current systems with a 58% increase in high needs block funding between 2014 and 2024 in response to increased demand. Despite this increased funding, there is no evidence of improved outcomes for CYP in the same ten-year period.

The National Audit Office report focuses on *how well* the system is delivering for CYP and their families, with the definition of SEN being *"broad, relative not absolute, and based on an assessment*"[12]p5]. The report identifies that the increased demand was linked primarily to increased identification of autism, speech, language and communication needs, and mental health issues. **Systemic challenges were identified and key to these was**

timely access to therapies. Recommendations include the need for system reform to enhance outcomes, establish financial sustainability, develop a shared understanding across sectors and create a long-term vision for inclusivity within mainstream education[12].

The Public Accounts Committee make strong, urgent recommendations to policy makers to better understand the increasing demand, raise the priority of CYP with SEND with Integrated Care Boards, and consider approaches that will support CYP earlier and in more inclusive mainstream settings[11].

This need for research to develop a needs-led model to inform workforce planning for equitable therapy services supporting CYP with SEND is strongly confirmed by the CYP and their families, to whom we have spoken. A parent of a young person with SEND reported, “Children and young people are experiencing trauma from needs not been met in a timely way”.

EXISTING EVIDENCE

Kessler and colleagues signpost to (i)issues with existing datasets and potential lines of enquiry, (ii)the need for more detailed exploration of therapy processes, and (iii)examples of issues of definition both in terms of types of need, metrics such as waiting list data, and workforce in their scoping study, “The Demand and Supply of Therapists for Children and Young People with Special Educational Needs and Disabilities”[13]. The review identifies the Balanced System® as a model is attempting to systematically address these challenges in understanding the interface between population need and workforce sufficiency.

Models of service delivery to meet need

Gascoigne (2006) introduced the concept of universal support for all CYP, targeted support for those at risk or with frequent needs, and specialist support for those requiring individualised, expert intervention in the Royal College of Speech and Language Therapists’ Position Paper[14]. The Royal College of Occupational Therapists has also endorsed this continuum in its guidance on working with CYP. Both organisations advocate for a structured, needs-based approach to delivering professional support[15]. A survey to establish a benchmark of current OT practice and approaches yielded over 400 responses from children’s OTs with 56% reporting to have a universal offer, 63% a targeted offer, and 93% delivering specialist input (*Gimeno, manuscript under preparation)[16]*. Evaluation of ‘tiered approaches’ is also emerging in physiotherapy[17].

A recent systematic mapping of school based allied health interventions for CYP affected by neuro-disability providing an evidence map of intervention approaches concludes, “*In the United Kingdom, future research should take forward school-based allied health interventions that relate directly to agreed research priorities. Internationally, future priorities include implementation of tiered intervention models and appropriate preparation and deployment of the education workforce”[18]*.

The Balanced System®

The Balanced System is a population-based, outcomes focused approach, integrating systems to support CYP with SEND, underpinned by a socio-ecological model. The approach seeks to understand the risk factors in a population in relation to the need for therapy and is not driven by diagnostic categories[19, 20]. The high-level outcomes framework across family and young person support, environments, workforce, identification and intervention is referenced in the *Best Start in Speech and Language: Guidance* for commissioners as an example of a useful system outcomes framework[21].

Whilst the framework originated in response to the challenges of supporting CYP with speech, language and communication needs, it has been extended to include occupational and physical needs. The population needs approach has also been applied to all three areas of need. However, the evidence-base for understanding population risk factors specific to occupational needs and physical needs has not been synthesised in a comparable way to the evidence of factors impacting development of speech, language and communication.

Currie *et al’s* report (2024) “*Delivery of services for Children and Young People with Special Educational Needs”*commissioned by the Department for Education, includes the Balanced System as one of five approaches identified for a deep dive review with local areas[22]. The report presents examples of impact on access to therapy services with reduced waiting times for support in for speech and language, but also occupational and physical needs,

“The long-term impact of this approach can be seen in data from Worcestershire where the implementation of the easy access drop-in has been maintained over time. Prior to implementation in 2011, the percentage of those waiting for initial appointments over 18 weeks was 26%. By 2013, this had dropped to 2% and has remained at that level or below ever since with those being outliers where other circumstances prevented access”[22]

Similar trends have been seen in occupational therapy and physiotherapy services in other areas.

The report also provides examples from participants about the evidence of the benefits of therapists working at place alongside education and early years colleagues,

"Using place-based support instead of clinics was described as tackling inequalities as children and young people with highest need are least likely to be brought into clinics,

therefore "if you go to where the child is, then you are reducing that inequality"[22]

The MOdelling Therapies for Improved Futures (MOTIF) study aims to provide workforce modelling that estimates the workforce needed to meet the needs of a population of CYP including those with SEND using the Balanced System approach.

Therapies in this proposal refer to: Occupational Therapy, Physiotherapy, and Speech and Language Therapy for CYP with SEND.

What is the research question / aims and objectives?

Research Question:

What is the estimated therapy workforce needed to deliver impactful, effective and cost-efficient support to meet the needs of CYP (0-18) with SEND?

Aim:

To inform policy, practice, and workforce planning for therapy services supporting CYP with SEND based on a needs-led model, thereby informing commissioning of provision.

Objectives:

1. Examine key risk factors at individual, environmental, social and economic levels that contribute to the needs of CYP with SEND for therapy services. ***Work package (WP-1)***
2. Review the policy and strategic guidance relevant to the funding, demand and supply of the therapy workforce to meet the needs of CYP with SEND (***WP-1)***
3. Gather the views regarding current therapy practices and their effectiveness, and the acceptability of a systems-based delivery model from key stakeholders (***WP-2)***
4. Identify the approaches, mechanisms, enablers and challenges for commissioning of therapies for CYP with SEND by health (including justice), education, and care commissioners in two selected regions: Greater Essex and Midlands (***WP-2)***
5. Generate therapies workforce models based on population need building on existing data and methods from a systems model (the Balanced System®) enhanced and further developed through this research programme. (***WP-3)***

RESEARCH PLAN AND METHODS

TEN KEY WORDS

SEND, population need, Speech and Language Therapy, Occupational Therapy, Physiotherapy, workforce, modelling, service delivery, integrated services, impact,

Research Plan:

Our public co-applicant (Hill) (a young person with lived experience of SEND and therapies) and parents/carers of CYP with SEND have been involved in the development of this research plan. Our public co-applicant and PPI groups will support the design, conduct and dissemination of all work packages.

Given the funding envelope and timeline, the study adopts two regions as exemplars to develop and test the workforce model, with further validation planned in future studies. Our proposed case study sites within Greater Essex and the Midlands regions cover a wide range of socio-economic and ethnic diversity, as well as rural areas. Tendring in Essex is ranked 36th out of 317 districts in England placing it in the most disadvantaged quintile in England based on the Indices of Multiple Deprivation and in Birmingham, which ranks 9th out of 317 as a Local Authority, 77% of wards are in the most disadvantaged quintile in England[23]. The 2021 Census reported 51.4% of the population as identifying as belonging to Black, Asian, or minority Ethnic (BAME) groups with over 90 different main languages spoken, Urdu, Punjabi and Bengali being the three most often reported. In Essex and

Worcestershire, the percentage of the population identifying as belonging to BAME groups is significantly lower at 7.6% and 4% respectively. However, 5.4% of the population in Essex and 3.4% of the population of Worcestershire identify as being of other European heritage with these communities focused in specific areas of both counties[24].

The MOTIF study will be delivered in three work packages (WPs) linked to five objectives:

WP1: Evidence synthesis (Objectives 1-2)

WP1.1: Scoping Review on risk factors that determine level of population need for support in CYP with SEND,

WP1.2: Scoping Review of SEND policy documents to understand policy context factors impacting on workforce modelling and therapies delivery.

WP2: Multi-Stakeholder Engagement (Objectives 3-4)

WP2.1: NATIONAL survey with therapy professionals.

WP2.2: NATIONAL focus group with professional associations.

WP2.3: REGIONAL* participatory workshops with CYP and families.

WP2.4: REGIONAL* focus groups with commissioners.

WP3: Modelling of therapy workforce (Objective 5).

WP3.1: NATIONAL analysis of population need for CYP needs.

WP3.2: REGIONAL* evaluation of whole system needs and delivery in systems WP3.3: REGIONAL* therapy workforce model applied in systems.

*REGIONAL refers to Midlands and Greater Essex

WP1: Scoping Reviews exploring population need and policy context in CYP with SEND *(months 1-8)*

Design: Scoping Reviews to (i) identify key epidemiological risk factors at individual, environmental, social and economic levels that drive the needs of the population (WP1.1) and (ii) analyse SEND policy documents to understand policy context factors impacting on workforce modelling and delivery of therapies for CYP with SEND (WP1.2).

WP1.1 Scoping Review of Epidemiological studies on risk factors that determine the needs of CYP with SEND.

Objective 1: Examine key risk factors at individual, environmental, social and economic levels that contribute to the needs of CYP with SEND for therapy services.

Methodology: Protocol based on the Arksey and O'Malley framework [25], co-with produced with our PPI and advisory groups through regular meetings and communications throughout.

The review will explore the evidence from peer reviewed publications, clinical guidelines, reports, and other grey literature up to 2025 reporting epidemiological risk factors for CYP within the diagnostic categories most associated with SEND. Further, we will search the literature for studies specifically relating to health inequalities and demographic factors associated with the range of therapy needs that CYP with SEND might have.

We will use Generative AI to support the scoping review by drafting research questions, summarising literature, generating search strategies, and identifying key themes to enhance efficiency and structure; all outputs will be reviewed by experts to ensure accuracy and relevance. Two public contributors will join the review team as co-researchers. We will register the protocol with Open Science Framework. COVIDENCE software will be used for screening/data extraction.

Search Strategy: MeSH Terms and Keywords will be used with the following main electronic databases: MEDLINE, EMBASE, Cochrane, CINAHL, PsycINFO, AMED.[26]

Reporting: Preliminary findings will be presented to PPI groups, and the discussion and views recorded and reflected in the final data synthesis and review report. Findings will be reported narratively, supported by tables,

following The Preferred Reporting Items for Systematic Reviews and Meta-Analysis Extension for scoping reviews (PRISMA-SCR)[27]. https://pubmed.ncbi.nlm.nih.gov/30178033/

WP1.2 Scoping Review of England Policy Context and Strategic guidance relevant to CYP with SEND

Objective 2: Review the policy and strategic guidance relevant to the funding, demand and supply of the therapy workforce to meet the needs of CYP with SEND.

Methodology:

A review of national level legislative, policy and strategic guidance relating to meeting the needs of CYP with SEND by HM Government, DfE, DHSC, Ministry of Justice, professional body guidance to members, Independent Associations guidance to members, Independent Providers of Special Education Advice guidance, Local Government Authority. We will include operational guidance from Local Authorities, the relevant Integrated Care System and NHS provider organisations within the two focus geographies to explore regional interpretations of the statutory guidance.

As per previous review, we will register the Arksey and O'Malley informed protocol with Open Science Framework and use the same reporting methodology guidelines. Generative AI as described above will support this scoping review. Our PPI groups will support the development of the review questions and protocol.

Study selection: National and regional policies, NHS and Integrated Care Systems strategies (e.g., workforce planning, therapy provision), local authorities SEND strategies, statutory guidance (e.g., SEND Code of Practice, NICE guidelines) will be reviewed from 2015 to current time.

Data Extraction: Policy documents will be analysed thematically. We will develop a matrix to extract and compare information systematically. Some suggested coding dimensions include (i)policy intent and goals, (ii)target population, (iii)workforce planning, (iv)supply factors, (v)demand estimation, (vi)funding and commissioning, (vii)implementation barriers.

Outputs:

WP1-1 will summarise available epidemiological information and enable the datasets currently curated in the Balanced System to be refined and extended to estimate population need for a broader group of CYP. This will help forecast future service requirements based on demographic trends, socioeconomic factors, and prevalence of conditions (WP3).

WP1-2will summarise the contextual policy framework for the therapy workforce meeting the needs of CYP with SEND. This will provide evidence of the strategic frameworks, policies, commissioning models and funding mechanisms at national and regional levels, that impact on the availability and distribution of the therapy workforce. This contextual analysis will inform evaluation of workforce models in WP3 including effectiveness of early intervention and prevention.

WP2: Multi-Stakeholder Engagement to identify current commissioning of therapy services, views and perspectives of what is working well within current provision models, and the acceptability of the Balance Systems Model *(months 3-18)*

WP2 will deliver objectives 3 ad 4 through four studies, two at a national level and two in the Midlands and Greater Essex regional areas in which the detailed modelling will be applied in WP3.

Objective 3: Gather the views regarding current therapy practices and their effectiveness, and the acceptability of a systems-based delivery model from key stakeholders: (i)therapy workforce, (ii)professional associations and (iii)CYP and Families (***WP-2.1, 2.2, 2.3)***

<u>WP-2.1: NATIONAL Professionals Survey</u>

Sampling and sample size: The estimated therapist membership of relevant professional bodies is 20-24,000 children's therapists. Using a 95% confidence level with a ±5% margin of error we will need a total of n=378 responses. For a ±3% margin of error we estimate 1,022 responses will be required. We will aim to recruit n=377 per therapy group. This is a realistic number based on our previous experience (Gimeno, survey of children's OTs with n=504 responses, *manuscript under preparation).*

Inclusion Criteria: Practising children's therapists (any context such as health, social care, local authorities, or independent practice).

Recruitment: via local clinical leaders, national professional groups and experts (s*ee collaborations*), via social media, sending reminders to non-respondents 1, 3, and 5 weeks after sending introductory emails.

Data Collection: Questionnaire informed by the Theoretical Framework for Acceptability[28] with a combination of Likert and open and closed questions, designed using the Checklist for Reporting Results of Internet E-surveys (CHERRIES)[29]. Data collection uses Qualtrics® online platform. Participants will complete the survey via an anonymous link or via a QR code. Survey questions will be co-produced with our PPI groups.

Areas of exploration on surveys will include:

1. Nature, place, type and duration of current practice
2. Perceived effectiveness of current practices
3. Evidence of impact measurement at CYP, family and wider contextual levels.
4. Acceptability of an integrated workforce working in a whole systems model.

Analysis: Closed and Likert scale questions will be analysed using descriptive statistics of frequencies and percentages as appropriate. Open-text responses will be analysed using content analysis.

To assess the validity and reliability of the survey, a confirmatory factor analysis and reliability analysis will be performed. A series of one-way Univariate Analyses of Variance (ANOVAs) will be conducted to explore possible differences between professional groups. Qualitative data will be analysed using directed content analysis. Our PPI groups will support the analysis of comments and interpretation of quantitative findings.

Output: Feedback aggregates and analyses the perspectives of therapists. This will provide a comprehensive view of current and potential models of delivery in context informing both the CYP workshops (WP2.4) and the workforce modelling with contextual variables (WP3.3).

WP 2.2 NATIONAL focus group for professional associations

Methodology and methods:

These qualitative studies (WP2.2 and WP2.4), will use online focus groups[30]; these are a pragmatic, cost-effective method to gather views of participants across the UK and have been found to be effective and appropriate[31].

At a national level, we will seek perspectives of professional associations about the following areas impacting CYP with SEND: (i) impactful support from therapies (ii) acceptability of a whole system approach, (iii) factors influencing workforce sufficiency of registered therapists iv) the role of assistant practitioners of all disciplines, v) key performanceindicators used to evaluate outcomes of therapies contribution to CYP outcomes, (vi) levels of satisfaction current workforce planning for therapies and the wider workforce, and (vii) facilitators and barriers to effective workforce planning.

The focus groups will facilitate discussion within professional associations representing all parts of the workforce supporting CYP with SEND, as well as individual contributions in response to direct questions providing a breadth of perspectives[32, 33].

Sampling and Sample Size: Purposive sampling through links to relevant professional associations. We will aim to conduct one focus group, containing approximately eight participants.

Inclusion Criteria: Representatives from key professional associations relevant to this research including, Royal College of Speech and Language Therapists, Royal College of Occupational Therapists, Chartered Society of Physiotherapy, National Association of Head Teachers, National Association for Special Educational Needs, Association of Directors of Children's Services, Institute of Health Visiting, and the Early Years Alliance.

Recruitment

Representatives will be invited via contact with the Chief Executive of each organisation asking them to nominate a suitable representative. The research team have working contacts with these organisations (*see letters of support)* and interest in this subject is predicted to enable recruitment.

Data collection: Focus groups via MS TEAMS using researchers' university account. Focus groups will be led by an experienced research team member with an assistant facilitator. Participants will be encouraged to ask questions.

The focus group topic guide will be developed by the research team, with extensive research experience in qualitative methods, and a comprehensive understanding of the CYP workforce (Gimeno, Gascoigne), our public co-applicant and PPI groups.

Analysis: Focus groups will be audio-recorded and transcribed verbatim, anonymised at the point of transcription, with unique IDs for individual participants. Data will be thematically analysed [34, 35] to identify emergent recurring and salient themes and any divergent views. The QMUL RA will lead the analysis, with support from Gimeno, Gascoigne, Tuppeny and Darby. Key themes will be agreed by discussion with research team. Codes will be compared and discussed until consensus is reached. Data management software system (NVIVO-11) will be used to facilitate data management and analysis. Our PPI groups will support the interpretation of findings from the focus groups, with support from Hill (public co-applicant) and Pagnamenta.

Output: An understanding of the workforce challenges that professional associations face in supporting CYP with SEND with a focus on the interface of the therapy workforce and the wider workforce. The views of this group will inform the final modelling outputs for therapies working in the wider context of CYP's services.

WP-2.3: REGIONAL Participatory workshops with CYP and Families

Methodology and Methods: Participatory workshops appropriate to each participant group while offering choices to meet individual needs and preferences to promote inclusive engagement. The workshops are linked iteratively with the workforce model development in WP3. The first set of synchronous and asynchronous workshops will explore perceptions of what a good whole system service model would look like. We will use the five strands from the Balance System Model: (i) family and young person support, (ii) environment, (iii) workforce, (iv) identification, and (v) intervention to provide an outcomes framework through which to explore potential service solutions for CYP and their families.

Our PPI group member and co-chair of the National Parent Carer Forum stressed the importance of the involvement of parents, carers and CYP in this WP: *"it will be really important to demonstrate how families have shaped how the Balanced System is implemented locally, how local delivery is adapted to meet the needs of local families".*

The outputs from the first set of workshops will inform the workforce modelling in WP3.3 which will generate a workforce model that could deliver the offer developed the initial workshops. This workforce model will then be brought to the second set of workshops using case scenarios to illustrate different workforce options. These will be the focus of the second workshops and will then inform the final workforce modelling within WP3.3.

Sampling and Sample Size: Convenience and purposeful maximum variation sampling techniques will ensure that a broad range of families participate in the study including CYP and families from diverse communities and those who are regularly under-represented in research. Each workshop will include 10 to 15 participants, aiming to balance diversity and manageability. A Sampling matrix (representing a spread of age, diagnoses, ethnicity, socio-economic background) will be used. This will ensure a range of perspectives while allowing active engagement from all attendees[36].

Recruitment: Recruitment via the two regional areas selected. Our PPI groups will co-produce recruitment pathways for CYP and parents and accessible videos/visuals to support recruitment. PPI has highlighted the importance of parent/carer networks, schools, social media and snowballing and involving PPI group members who work with community groups to act as 'trusted advocates' to support recruitment and provide links with them to overcome barriers of trust and stigma around SEND. We will monitor recruitment of families for inclusivity and diversity and adapt our recruitment strategies as needed to recruit a sample that is representative (through purposive sampling) with respect to disadvantage and ethnicity. Our PPI work indicated that it will be important to include families who have experience of therapies and those who have not been able to access support to reflect current system and poor access to provision and diverse lived experiences.

Data Collection: Participatory workshops will be held with CYP with SEND and families in Greater Essex and Midlands. Our public co-applicant (Hill), PPI groups and representatives of the National Parent Carer Forums (Harrison and Lambe) will advise and support these workshops, including co-production of workshop materials.

Data will be collected through n=2-4 workshops at different times (m10 and m16-17). We will conduct 2 synchronous (1 face-to-face and 1 virtual via MS Teams) and 2 asynchronous (virtual Basecamp forum-based) workshops. A shared, secure, online space will also be available to participants for them to review workshop materials and comment, share ideas outside of the workshops, these will contribute to study data. As recommended by our PPI groups, we will gather information on communication supports and reasonable adjustments needed prior to workshops and provide interpreters (including BSL interpreters) to ensure accessibility.

Case scenarios will be provided drawing from the service mapping in WP3.1 and informed by the evidence from WP 2.1, 2.2 and 2.4. Participants will receive a preparation workshop pack which might include video/audio files to make it inclusive following advice from our PPI group.

We will explore transformational challenges, visualise a journey and record the pathway and work with CYP and families and other stakeholders iteratively to feed into the modelling.

Analysis: A directed content analysis using a deductive coding framework. Data will be analysed after each workshop to develop prototype Theory of Change maps which can be presented and iteratively refined. The PPI groups will support the analysis and development of the ToC.

Output: A multi-stakeholder feedback model that aggregates and analyses the perspectives of CYP, families, therapists, commissioners, and professional associations. This model can highlight the most effective therapy supports and will feed directly into the workforce modelling in WP3.3.

WP2.4 REGIONAL focus groups for commissioners Objective 4. Identify the approaches, mechanisms, enablers and challenges for commissioning of therapies for CYP with SEND by health (including justice), education, and care commissioners in the two selected regions: Greater Essex and Midlands.

Methodology: Online focus groups to gain commissioners' views on(i) nature and type of commissioning, (ii) financial envelope of commissioning, (iii) key performance indicators used to evaluate success, (iv) levels of satisfaction with impact and outcomes of current commissioning, and (v) facilitators and barriers to effective commissioning.

Sampling and Sample Size: Purposive sampling to identify commissioners across the different commissioning agencies. We will aim to conduct between three to six focus groups, containing 6-8 participants each, giving a potential maximum sample of 48. This number is in line with a systematic review of sample sizes required in qualitative research. The sufficiency of the data will drive the final sample size[37, 38]

Inclusion Criteria: Commissioners in Health (primarily Integrated Care Systems but also NHS England specialist commissioning and Justice) and Local Authority (including Public Health, SEND and Inclusion and Social Care), who are responsible for commissioning the therapy workforce in the chosen regions. Representation from Headteachers who commission these therapies directly in each of the areas would be included.

Recruitment: Commissioners will be recruited from chosen geographies, with the support of Integrated Care Board Children's Leads/Local Authority SEND Leads. The research team have working contacts within both geographies enabling this recruitment.

Data collection and Analysis will follow same methodology as WP2.2

Output: A comprehensive map of the commissioning processes across health, education, and care sectors in two selected geographical regions. This map will identify gaps and overlaps in commissioning of service provision and inform the acceptability of the workforce modelling approaches in WP3.3

WP3 Modelling therapy workforce to inform policy (*months 0-20***)**

Objective 5: Generate therapies workforce models based on population need building on existing data and methods from a systems model (the Balanced System®) enhanced and further developed through this research programme.

Building the workforce model requires an iterative process:

WP3.1 CYP needs analysis: Estimation and assessment of the levels of need within a geographically defined population of CYP.

WP 3.2 Systems needs analysis: Analysis of the contextual landscape of services and the broader system offer within the selected geography

WP3.3 Modelling the therapy workforce: Development of therapy workforce models that are responsive to the identified needs and contextual factors.

WP3.1 CYP needs analysis *(Months 1-12)*: We will estimate levels of need among CYP within defined geographic populations, using data-driven approaches to identify the scale, distribution, and nature of need to inform the support required. A detailed demographic analysis will be conducted by ward, neighbourhood, and age band, by developing an enhanced version of the Balanced System approach. This will include indicators of disadvantage such as the Index of Multiple Deprivation, Income Deprivation Affecting Children Index, Social Mobility Index, and Her Majesty's Revenue and Customs data at the Lower Layer Super Output Area level. Contextual factors, including rurality and population diversity, will also be considered. This analysis will be complemented by the integration of SEND data at

both local authority and school levels, alongside educational attainment data for the Early Years Foundation Stage, Key Stage 2, and Key Stage 4 for CYP with SEND. Our PPI groups will provide guidance on further contextual factors. Predictive modelling techniques[39] will estimate the prevalence of Speech, Language and Communication Needs, as well as needs related to occupation, participation, and physical development, at ward and neighbourhood levels. The Balance System will serve as the central platform for curating and integrating these datasets, incorporating relevant findings and resources identified in WP1 (CYP Needs), including additional epidemiological factors associated with Speech and Language and communication needs beyond those already considered. WP1 will also identify prevalence data for SEND categories that impact these domains, ensuring a more comprehensive understanding of need across functional areas. Automation will support the creation of standardised datasets for each area, enabling seamless integration into the Balance System. To reduce bias and ensure inclusivity and equity in our predictive models, we will follow established guidelines such as the BE-FAIR Equity Framework[40].

Outputs: Updated and refined database to support workforce modelling. Comprehensive map outlining population-level predicted needs at the national scale.

WP 3.2 System Needs Analysis *(M01–M15)***:**

Methodology: We will conduct a system needs analysis across three Integrated Care Systems, Local Authorities, and NHS provider networks in two regions: Greater Essex (comprising Southend, Essex, and Thurrock) and parts of the Midlands (Herefordshire, Worcestershire, and Birmingham), plus one Midlands area not currently engaged in a whole systems approach (Nottinghamshire). These areas were selected for their diverse community contexts—coastal, rural, and urban settings—and varied linguistic, ethnic, cultural, and socio-economic profiles.

Data collection: In each case study area, we will collect quantitative and qualitative data on service provision and system structures to create a detailed contextual map. PPI groups will guide data collection. Data collection will be embedded within the Balance System tool to ensure consistency and interoperability across sites. We will collect workforce data from provider organisations such as the number of substantively funded whole-time equivalent therapists, assistant practitioners, and apprentices, broken down by Agenda for Change band, funding source, and specialist clinical roles.

Data analysis: Data will be analysed against population size and predicted level of need. Additional data will include vacancy rates, staff turnover, referrals, discharges (over a 12-month period), waiting times, and active caseloads - broken down by age band, support setting, and SEND status (no SEND, SEN Support, or Education, Health and Care Plan). Outcome measures routinely collected by services will also be included. Qualitative text data from the provision mapping exercise will be analysed using advanced AI techniques, including natural language processing, topic modelling, sentiment analysis, and automated summarisation to identify key themes, service gaps, and stakeholder perspectives[41]. This analysis will inform strategic planning and workforce modelling, supported by findings from WP2 (System Needs). All findings will be validated through engagement with PPI groups and relevant stakeholders to ensure accuracy, relevance, and contextual alignment.

Outputs: Contextual descriptions of service provision to inform modelling activity in WP3.3. Generative AI powered qualitative data analysis methodology will be implemented and functioning.

WP3.3 Modelling the therapy workforce *(Months 6-24)*

Methodology: Development of therapy workforce models based on the needs identified in WP3.1 and the local system context established in WP3.2. The aim is to ensure the appropriate mix, capacity, and distribution of professionals to deliver effective and equitable support to CYP. We will enhance the Balanced System with new analytical tools, including explainable AI[42] and feedback features to support dynamic, data-informed workforce planning. The models will estimate workforce requirements based on current and predicted levels of need, local service structures, and resource availability. By applying different contextual scenarios - shaped by our PPI group - the models will allow comparison of workforce configurations and outcomes. This will support strategic decision-making by illustrating how variations in system design and resource allocation may impact service delivery and effectiveness.

Output: Workforce model. First workforce models implemented and tested in test sites

Dissemination, outputs and Impact

A dissemination and communication plan will be developed and implemented at the start of the project and will be an agenda item on all project and steering group meetings. The objectives of this plan will be to: 1) raise awareness of MOTIF and its activities among the research community, professionals, practitioners, commissioners, policymakers, and the public; 2) share the project findings with stakeholders at regional, and national levels, via accessible reports, policy briefs, presentations, press releases and targeted publications; 3) facilitate uptake and further refinements of

the model to ensure an appropriate therapy workforce to serve the needs of CYP with SEND.

A broad-based dissemination approach will ensure communication and dissemination of project findings to all stakeholders:

- To academic audiences through Open Access academic publications and conferences,
- To professional, practitioner, policy and wider stakeholder audiences through papers in selected professional publications, policy briefs, press releases, conferences, podcasts,
- Targeted networking events and bespoke workshops,
- A project website will be set up to keep stakeholders abreast of the progress of the study,
- Social media outlets will be key avenues for communication and dissemination. Our public co-applicant and the PPI groups will support the dissemination of the research, inform key audiences and effective ways of engaging these in dissemination activities and co-produce easily accessible research summaries and a public-facing film/podcast. The group will also support dissemination to managers and commissioners.

The use of accessible language for all public-facing documents was highlighted as critical through our pre-application PPI work.

The final outputs of this research will be a series of reports including,

1. We anticipate workforce predictions relevant to the current population of CYP with SEND in the context of current service delivery and predictive models based on future population predictions and the development of whole system service integration at place.
2. Workforce modelling reports that predict the potential workforce requirements from children's therapies working as part of the whole system to meet the needs of children and young people with SEND in selected Integrated Care Systems within Greater Essex and Midlands

There will be interim reports for each of the work packages.

STUDY TIMELINE:

Start date: 01/03/2026/ Study duration: 24 months from main contract signing, including set-up phase for collaboration agreements, and local research governance approvals.

Months 1-12: WP1 scoping reviews, ethics application for WP2 (HRA ethics for stakeholders workshops), QM ethics for surveys (Studies 2+3 in WP2), survey data collection, modelling of current available data on Balance System from Gascoigne, inclusion of additional datasets and evidence arising from WP1.1, collection of service level data from sites(WP3)

Months 13-18: WP2 Qualitative interviews recruitment, data collection and analysis. Analysis of surveys. WP3 inclusion of service level data and contextual place data in modelling, generation of first models for use in WP2 workshops

Months 18-24: Refining models with feedback from WP2 workshops. Final modelling of sites.

PROJECT MANAGEMENT

The two Co-PIs will oversee and manage the project at an operational and governance level, ensuring all deliverables are met within the timeline. They will co-ordinate and manage the project team and project steering group. All co-applicants will make up the ***project management*** team with responsibility for operationalising the study, meeting monthly with Research Assistants to ensure the smooth running of the project.

There will be a project ***steering group*** comprising independent members who will provide oversight and guidance to the project team and who will meet twice a year. This will be comprised of key stakeholders and people in strategic roles in commissioning and delivering services, and in policy, who can guide and support future dissemination. The group will meet four times during the project.

The ***PPI Groups*** (separate groups of 8 parents/carers to meet online, and two groups of 4 CYP with SEND to meet face to face in each region (8 in total)) as advised by our PPI work in the preparation of this grant) will feed into the project management team and the steering group. The groups will meet six times over the two years aligned with key stages in each work package and steering group meetings. We will ensure we reach out to CYP and families from minority ethnicities, including non-English speaking families via local parent carer forums in each region, which have strong networks with diverse communities. We will use the NIHR INCLUDE Ethnicity Framework[43](https://www.trialforge.org/trial-forge-centre/include/ to co-produce an EDI strategy with our PPI groups to ensure embedded equality, diversity, and inclusion at all levels.

ETHICS

Ethical approval will be gathered through Queen Mary University of London for the focus groups and survey of professionals as recruitment and data collection will not be conducted via the NHS so no HRA review is required for this element.

An ethics application will be made for HRA and NHS Research Ethics Committee review and favourable opinion for all other aspects of the study, along with Trust Capacity and Compliance Checks.

ROLES AND RESPONSIBILITIES

Gascoigne: *Operational lead of the project. Day to day project management, ethics submission, patient identification, recruitment including consenting, data collection, analysis, model development, outputs*

Joffe: *Co-lead of the project. Support for lead applicant with project management, project governance, ethics submission, PPI, methodological expertise, data collection support, analysis, model development, outputs.*

Gimeno: *Co-applicant. Oversight and support for QMUL based research assistant, ethics submission, PPI, methodological expertise, data collection support, analysis, model development, outputs*

Pagnamenta: *Co-applicant. PPI Lead, ethics submission, PPI strategy, PPI groups and training, PPI evaluation and monitoring*

Reinhold: *Co-applicant. Use of AI, building simulation for modelling*

Tuppeny: *Co-applicant. Leading on data collection from an Occupational Therapy Perspective in WP3*

Darby*: Co-applicant. Leading on data collection in WP3 from a physiotherapy perspective*

Hill: P*ublic co-applicant with lived experience of SEND and therapies, co-PPI lead, PPI groups, PPI strategy**,** support design and delivery of workshops with families, output*

EXPERTISE

Large scale complex project management: Gascoigne, Joffe, Gimeno, Tuppeny

Workforce planning/needs analysis: Gascoigne, Joffe, Tuppeny, Darby

Clinical and Management experience working with CYP with SEND and families across education and health: Gascoigne, Joffe, Pagnamenta, Tuppeny, Darby, Gimeno ***Methodology expertise:*** Gimeno, Joffe, Pagnamenta**,** Scherer

Children's research: Gimeno, Joffe, Pagnamenta

PPI with CYP: Pagnamenta, Gimeno, Joffe

Policy: Gascoigne, Joffe, Tuppeny

Artificial Intelligence, and statistical modelling; Scherer

Needs analysis, service design and impact measurement in complex systems: Gascoigne, Tuppeny, Darby.

SUCCESS CRITERIA AND BARRIERS TO PROPOSED WORK

Success criteria will be meeting the objectives for the study in accordance with the timeline. Potential risks to the study and mitigation plans are as follows:

Risk 1: Delay on R&D/ethics approval

Mitigating plans:

- QMUL Ethics submission for online professional survey and for focus groups with commissioners and professional colleges. Low risk applications approval in <2-3 weeks.
- NHS REC via HRA application for parent survey, and qualitative study of CYP and Families in parallel to gathering survey data.

Risk 2: Inadequate Recruitment

Mitigating plans:

- Regular contact, training, and upskilling
- Engage PPI team to identify solutions
- Use of social media
- Recruitment to be done through the research team (all co-applicants and collaborators)

Risk 3: Delay in recruitment of research assistants

Mitigating Plans:

- For the QMUL based RA, a post-doc qualitative research associate has been identified from Gimeno's lab and will be ready to start in the project minimising delays with set up.
- Gascoigne team already able to identify RA to commence in January 2026

Risk 4: Delay in data collection

Mitigating Plans:

- Ethics applications in parallel for studies in WP2
- Engage PPI team to identify solutions
- Recruit research interns from Gimeno awarded NIHR North London Health and Care Professionals research internships/ other regions in England

Risk 5: Milestones unmet

Mitigating plans:

- Monthly review in PMG
- Open up opportunities for research experience through internships (Gimeno)

	Pre-award	Y1 (2026)												Y2 (2027)											
		Q1			Q2			Q3			Q4			Q1			Q2			Q3			Q4		
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
OVERALL PROJECT MANAGEMENT																									
Recruitment: Research Assistants/PPI group/Techinician																									
Contracts CIC and HEIs																									
Website set up																									
Balance System database preparation																									
PPI meetings - parent/carer group (online)																									
PPI meetings - CYP group (face to face)																									
PPI training																									
Steeering group																									
PROJECT PLAN																									
WP1 - Scoping Reviews																									
WP1.1 Scoping review of risk factors																									
Write up protocol																									
Searches in 3 databases																									
Screening abstracts and full textx																									
Extraction of Data																									
Write up results																									
Inform modelling																									
WP1.2 Scoping Review of Policy Context																									
Write up protocol																									
Searches of policy documents																									
AI analysis of documents																									
Verification of AI analysis																									
Write up results to inform focus groups with commissioners																									
WP2: Multi-modal Stakeholder																									
WP2.1: NATIONAL Survey of Professionals																									
Write up protocol																									
Ethics paperwork (QM) (application, survey, PIS, Consent)																									
Data Collection																									
Survey clean up of results																									
Survey analysis																									
Feedback results to modelling																									
WP2.2 NATIONAL Focus Group with Professional Associations																									
Write up protocol																									
Preparation of case scenarios/materials for the groups																									
Ethics paperwork (QM) (application, PIS, Consent)																									
Recruitment of participants																									
Focus group online																									
Transcription of focus group																									
Analysis of focus group																									
Feedback results to modelling																									
WP2.3: REGIONAL Workshops of stakeholders																									
Case Scenarios preparation																									
Write up protocol																									
Ethics paperwork (HRA application, PIS, consent…)																									
Data collection (multi-method of workshops)																									
Data analysis and reporting																									
Feedback to modelling																									
WP2.4 REGIONAL Focus Groups with Comissioners																									
Write up protocol																									
Ethics paperwork (QM) (application, PIS, Consent)																									
Recruitment of Commissioners																									
Focus groups online																									
Transcription of focus groups																									
Analysis of focus groups																									
Feedback results to modelling																									
WP3. MODELLING																									
WP3.1 Modelling population need																									
Review existing Balanced System datasets and algorythm. Preparing for integration of new evidence from WP1.1																									
Data integration from WP1.1 to produce prediction of need																									
WP3.2 Systems needs analysis in two regions																									
Populating the Balanced System tools with workforce, caseload, waiting list data from participating services (quantitative) & service and wider system mapping (qualitative)																									
Generative AI methodology implemented and functioning																									
Outputs of commissioning focus groups (2.4), practitioner survey (2.1) and initial stakeholder workshops (2.3) included in contextual variables																									
AI driven analysis of system factors & Contextual descriptions complete to inform workforce modelling in 3.3																									
WP3.3 Workforce modelling in two regions																									
Model development and testing iteratively across stakeholder workshops																									
Final workforce model																									

DISSEMINATION																							
Public-facing research summaries																						★	
Public-facing film/podcast/animation																							

7. Patient & Public Involvement

Please describe how patients and the public have been involved in developing this proposal.

Routes for recruitment have been added: parent support groups/networks, particularly working with national parent carer networks to reach diverse communities and local organisations already working with CYP with SEND, schools, social media and snowballing. It will be important to include parents/carers who have experience of therapies and those who have not been able to access support to reflect diverse lived experiences. We have included 'trusted advocates' to support recruitment and data collection with diverse communities in recognition that trust and stigma around SEND are likely to be barriers to participation for some communities and CYP. We will use interpreters and translation for workshop materials as needed (heritage languages and/or BSL). Vouchers have been included as incentives for parents/carers and CYP.

Parents/carers and CYP with SEND have been added as an important participant group to be included in WP2. A range of approaches, including workshops and written contributions with easy read information, verbal explanations and interpreters (including BSL) were recommended to ensure inclusivity of families with different experiences who are experiencing difficult journeys, some will have disabilities/communication needs. The involvement of organisations experienced in hosting public involvement work has been added to support engagement with workshops (WP2), and organisations skilled in supporting the participation of CYP in each region to support our PPI work.

Gathering information on communication supports and reasonable adjustments needed prior to workshops and asking CYP if they would like parents/carers or a trusted adult to be included have all been added.

It was felt important that parents/carers and CYP participating in the WPs had very recent or current experience of therapies/trying to access therapies.

The PPI groups will be involved throughout in supporting the design, conduct and dissemination of the research. The use of accessible language for all public-facing documents was highlighted as critical.

Please describe the ways in which patients and the public will be actively involved in the proposed research, including any training and support provided.

PPI groups will include 6 parents/carers of CYP with SEND including parents/carers from Essex and Midlands (including Victoria Riley, Dave Harford), and regional Parent Carer Forum directors for Essex and Midlands (Jo Harrison, Elaine Lambe). Parent Carer Forums will recruit 2 further parents/carers from diverse communities in these regions. Separate groups of 4 CYP with SEND will be recruited in each region and supported via local groups skilled in enabling participation e.g. Essex Multi-schools Academy. Our PPI work has highlighted over-recruitment will be required.

PPI groups will meet three times per year and work closely with our public co-applicant and EP to oversee and inform each WP:

WP1: Support development of review questions, protocols, interpretation of findings, final synthesis and report.

WP2: Co-produce recruitment pathways for parents/carers and CYP and accessible recruitment videos/visuals. Group members who work with community groups will act as 'trusted advocates' to reach wider communities. Co-produce survey questions (WP2.1), focus group topic guides (WP2.2, 2.4), and workshop design and materials (WP2.3) to ensure acceptability. Members from Parent Carer Forums will facilitate workshops. PPI groups will support data analysis and development of Theory of Change.

WP3: Inform contextual factors, collection of service provision data and contextual scenarios, and review and validate findings.

Dissemination: Inform key audiences and effective ways of engaging these in dissemination activities, co-produce easily accessible research summaries and a public-facing film/podcast and support dissemination to managers and commissioners.

The GRIPP2 checklist will be used to plan and monitor PPI.

All members will be renumerated in accordance with NIHR guidance, taking a flexible approach to account

for individual circumstances and ensuring that there are no negative financial impacts of participation. **Training and support: The PPI groups will receive an induction, training and support from EP, SH and the research team. Training and support will be provided by Parent Carer Forums on public involvement. Separate meetings will be offered with EP before/after PPI group meetings to ensure members have been able to communicate their views. CYP and parents will be supported to participate in meetings, using communication supports as needed, by EP (a specialist SLT).**

In rare cases where proposals do NOT involve patients and the public, clear justification must be provided.

8. Detailed Budget

Note: The SoECAT Form is included as a separate document.

Staff Costs

Budget Item 1

Staff Posts and Salaries

Staff name	Sophia Riley-Hill

Staff type	Lead & Co-applicant

Organisation name	PPI representative

Organisation type	Other

Role	Public co-applicant

Grade	not applicable

Salary (£)
41,240

On-costs (£)
0

Current annual salary (£)	£41,240.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£41,240.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	5	5	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£2,062.00	£2,062.00	£4,124.00

Budget Item 2

Staff Posts and Salaries

Staff name	Hortensia Gimeno

Staff type	Lead & Co-applicant

Organisation name	Queen Mary University of London

Organisation type	HEI (UK)

Role	Co-Applicant

Grade	Clinical Reader

Salary (£)
77,901

On-costs (£)
22,620

Current annual salary (£)	£100,521.00

Annual Costs of Staff Posts

Rate (%)	80

Base annual salary with rate	£80,416.80

	Year 1	Year 2	
The year 1 value should always be 0%.	0	1.65	

	Year 1	Year 2	
FTE	12	12	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£9,650.02	£9,809.24	£19,459.26

Budget Item 3

Staff Posts and Salaries

Staff name	QMUL PDRA

Staff type	Research Staff

Organisation name	Queen Mary University of London

Organisation type	HEI (UK)

Role	Post-Doctoral Research Associated

Grade	4RES

Salary (£)
45,974

On-costs (£)
13,042

Current annual salary (£)	£59,016.00

Annual Costs of Staff Posts

Rate (%)	80

Base annual salary with rate	£47,212.80

	Year 1	Year 2	
The year 1 value should always be 0%.	0	5.46	

	Year 1	Year 2	
FTE	60	60	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£28,327.68	£29,874.37	£58,202.05

Budget Item 4

Staff Posts and Salaries

Staff name	Emma Pagnamenta

Staff type	Lead & Co-applicant

Organisation name	University of Reading

Organisation type	HEI (UK)

Role	Co-applicant, PPI Lead

Grade	8

Salary (£)
66,242

On-costs (£)
19,060

Current annual salary (£)	£85,302.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£85,302.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	6.11	

	Year 1	Year 2	
FTE	10	10	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£8,530.20	£9,051.40	£17,581.60

Budget Item 5

Staff Posts and Salaries

Staff name	Prof Victoria Joffe

Staff type	Lead & Co-applicant

Organisation name	University of Essex

Organisation type	HEI (UK)

Role	Co-Lead Investigator

Grade	11

Salary (£)
109,679

On-costs (£)
32,154

Current annual salary (£)	£141,833.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£141,833.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	1.51	

	Year 1	Year 2	
FTE	10	10	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£14,183.30	£14,397.47	£28,580.77

Budget Item 6

Staff Posts and Salaries

Staff name	Prof Reinhold Scherer

Staff type	Lead & Co-applicant

Organisation name	University of Essex

Organisation type	HEI (UK)

Role	Co-applicant

Grade	11

Salary (£)
88,324

On-costs (£)
25,720

Current annual salary (£)	£114,044.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£114,044.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	1.88	

	Year 1	Year 2	
FTE	10	10	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£11,404.40	£11,618.80	£23,023.20

Budget Item 7

Staff Posts and Salaries

Staff name	Marie Gascoigne

Staff type	Lead & Co-applicant

Organisation name	BETTER COMMUNICATION CIC

Organisation type	Commercial (UK)

Role	Primary Investigator

Grade	Director

Salary (£)
100,000

On-costs (£)
20,000

Current annual salary (£)	£120,000.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£120,000.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	20	20	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£24,000.00	£24,000.00	£48,000.00

Budget Item 8

Staff Posts and Salaries

Staff name	Sharon Tuppeny

Staff type	Lead & Co-applicant

Organisation name	BETTER COMMUNICATION CIC

Organisation type	Commercial (UK)

Role	Co-applicant

Grade	Senior Associate

Salary (£)
68,631

On-costs (£)
19,902

Current annual salary (£)	£88,533.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£88,533.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	8	8	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£7,082.64	£7,082.64	£14,165.28

Budget Item 9

Staff Posts and Salaries

Staff name	Maxine Darby

Staff type	Lead & Co-applicant

Organisation name	BETTER COMMUNICATION CIC

Organisation type	Commercial (UK)

Role	Senior Associate

Grade	AfC 8b

Salary (£)
68,631

On-costs (£)
19,902

Current annual salary (£)	£88,533.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£88,533.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	5	5	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£4,426.65	£4,426.65	£8,853.30

Budget Item 10

Staff Posts and Salaries

Staff name	RA BCCIC

Staff type	Research Staff

Organisation name	BETTER COMMUNICATION CIC

Organisation type	Commercial (UK)

Role	RA

Grade	Associate

Salary (£)
40,824

On-costs (£)
10,838

Current annual salary (£)	£51,662.00

Annual Costs of Staff Posts

Rate (%)	100

Base annual salary with rate	£51,662.00

	Year 1	Year 2	
The year 1 value should always be 0%.	0	0	

	Year 1	Year 2	
FTE	60	60	

	Year 1	Year 2	
Months	12	12	

	Year 1	Year 2	Total
Rate calculation	£30,997.20	£30,997.20	£61,994.40

Travel, Subsistence & Conference Costs

Budget Item 1

Item description	Investigator travel

Type of Cost	Travel

Organisation Name All cost items must be assigned to the organisation who will be incurring the costs. When typing the organisation name, a drop down list will appear with a list of organisation names to select.
BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate (%)	100

	Year 1	Year 2	
Costs	£456.00	£456.00	

	Year 1	Year 2	Total
Total rate calculation	£456.00	£456.00	£912.00

Budget Item 2

Item description
Travel for people to attend workshops (including people with reduced mobility, children in wheelchairs)

Type of Cost	Travel

Organisation Name All cost items must be assigned to the organisation who will be incurring the costs. When typing the organisation name, a drop down list will appear with a list of organisation names to select.
BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£1,120.00	£1,120.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£1,120.00	£1,120.00	£2,240.00

Budget Item 3

Item description	Conference attendance

Type of Cost	Conference

Organisation Name All cost items must be assigned to the organisation who will be incurring the costs. When typing the organisation name, a drop down list will appear with a list of organisation names to select.
BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£3,000.00	£3,200.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£3,000.00	£3,200.00	£6,200.00

Budget Item 4

Item description	Workshop attendees - subsistence

Type of Cost	Subsistence

Organisation Name All cost items must be assigned to the organisation who will be incurring the costs. When typing the organisation name, a drop down list will appear with a list of organisation names to select.
BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£450.00	£450.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£450.00	£450.00	£900.00

Equipment Costs

Budget Item 1

Item description	Encrypted Recorder

Organisation Name	Queen Mary University of London

Organisation Type	HEI (UK)

Rate %	80

	Year 1	**Year 2**	
Costs	£500.00	£0.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£400.00	£0.00	£400.00

Budget Item 2

Item description	RA (QMUL) laptop

Organisation Name	Queen Mary University of London

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs	£1,000.00	£0.00	

	Year 1	Year 2	Total
Total rate calculation	£800.00	£0.00	£800.00

Consumables

Budget Item 1

Item description
Materials for activity based interventions (pens, stickers, printing)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£450.00	£0.00	

	Year 1	Year 2	Total
Total rate calculation	£450.00	£0.00	£450.00

Budget Item 2

Item description	Materials for workshops

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£600.00	£600.00	

	Year 1	Year 2	Total
Total rate calculation	£600.00	£600.00	£1,200.00

Patient & Public Involvement, Engagement and Participation Costs

Budget Item 1

Item description	Attendance - PPI voucher reimbursement

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£900.00	£900.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£900.00	£900.00	£1,800.00

Budget Item 2

Item description	Preparation for meetings - voucher reimbursement

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£1,050.00	£1,050.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£1,050.00	£1,050.00	£2,100.00

Budget Item 3

Item description	Childcare costs (meetings)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£240.00	£240.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£240.00	£240.00	£480.00

Budget Item 4

Item description	Carer costs (meetings)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£30.00	£30.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£30.00	£30.00	£60.00

Budget Item 5

Item description	Refreshments

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£210.00	£210.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£210.00	£210.00	£420.00

Budget Item 6

Item description	Travel reimbursement for CYP PPI group

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£480.00	£480.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£480.00	£480.00	£960.00

Budget Item 7

Item description	Travel of PPI lead to CYP PPI meetings

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£300.00	£300.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£300.00	£300.00	£600.00

Budget Item 8

Item description
Travel of PPI public co-applicant plus parent to CYP meetings

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£450.00	£450.00	

	Year 1	Year 2	Total
Total rate calculation	£450.00	£450.00	£900.00

Budget Item 9

Item description
Travel for PPI group members - dissemination events

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs		£1,000.00	

	Year 1	Year 2	Total
Total rate calculation	£0.00	£1,000.00	£1,000.00

Budget Item 10

Item description	Training for PPI group

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£2,000.00		

	Year 1	Year 2	Total
Total rate calculation	£2,000.00	£0.00	£2,000.00

Budget Item 11

Item description
Talking Mats – Foundation Training + Consulting Children & Young People set

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£389.00		

	Year 1	**Year 2**	**Total**
Total rate calculation	£389.00	£0.00	£389.00

Budget Item 12

Item description
Attendance - National Parent Carer Forum representatives

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£300.00	£300.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£300.00	£300.00	£600.00

Budget Item 13

Item description	Meeting room costs (CYP group)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£600.00	£600.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£600.00	£600.00	£1,200.00

Budget Item 14

Item description	Facilitation of CYP PPI groups

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£300.00	£300.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£300.00	£300.00	£600.00

Other Direct Costs

Budget Item 1

Item description	Advisory group meetings

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£240.00	£240.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£240.00	£240.00	£480.00

Budget Item 2

Item description	Interpreters

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£640.00	£640.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£640.00	£640.00	£1,280.00

Budget Item 3

Item description	Transcription (focus groups)

Organisation Name	Queen Mary University of London

Organisation Type	HEI (UK)

Rate %	80

	Year 1	**Year 2**	
Costs	£1,728.00	£864.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£1,382.00	£691.00	£2,073.00

Budget Item 4

Item description	Participant incentives

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£1,500.00	£1,500.00	

	Year 1	Year 2	Total
Total rate calculation	£1,500.00	£1,500.00	£3,000.00

Budget Item 5

Item description
Voice recording/video recording for inclusive resources, i.e. participant information sheets

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£1,240.00	£1,240.00	

	Year 1	Year 2	Total
Total rate calculation	£1,240.00	£1,240.00	£2,480.00

Budget Item 6

Item description	Archiving costs

Organisation Name	Queen Mary University of London

Organisation Type	HEI (UK)

Rate %	80

	Year 1	Year 2	
Costs		£500.00	

	Year 1	Year 2	Total
Total rate calculation	£0.00	£400.00	£400.00

Budget Item 7

Item description	Covidence platform access

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£880.00	£0.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£880.00	£0.00	£880.00

Budget Item 8

Item description	Room booking for workshops (wheelchair accessible)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£800.00	£800.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£800.00	£800.00	£1,600.00

Dissemination Costs

Budget Item 1

Item description	Dissemination event

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£0.00	£2,400.00	

	Year 1	Year 2	Total
Total rate calculation	£0.00	£2,400.00	£2,400.00

Budget Item 2

Item description	Infographics (including for workshops)

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs	£840.00	£1,300.00	

	Year 1	Year 2	Total
Total rate calculation	£840.00	£1,300.00	£2,140.00

Budget Item 3

Item description	Co-produced film from workshops

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	Year 2	
Costs		£3,000.00	

	Year 1	Year 2	Total
Total rate calculation	£0.00	£3,000.00	£3,000.00

Budget Item 4

Item description	Study Website

Organisation Name	BETTER COMMUNICATION CIC

Organisation Type	Commercial (UK)

Rate %	100

	Year 1	**Year 2**	
Costs	£2,000.00	£0.00	

	Year 1	**Year 2**	**Total**
Total rate calculation	£2,000.00	£0.00	£2,000.00

Indirect Costs

Budget Item 1

Item Description	Estate Costs and Technicians (QMUL)

Organisation Name	Queen Mary University of London

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate (%)	80

	Year 1	**Year 2**	
Costs	£18,902.16	£18,902.16	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£15,122.00	£15,122.00	£30,244.00

Budget Item 2

Item Description	Indirect Costs (QMUL)

Organisation Name	Queen Mary University of London

Type of Cost	Other Indirect Costs

Organisation Type	HEI (UK)

Rate (%)	80

	Year 1	Year 2	
Costs	£54,383.76	£54,383.76	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£43,507.00	£43,507.00	£87,014.00

Budget Item 3

Item Description
Estate Costs and Technicians (University of Reading)

Organisation Name	University of Reading

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate (%)	100

	Year 1	Year 2	
Costs	£2,632.50	£2,632.50	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£2,633.00	£2,633.00	£5,266.00

Budget Item 4

Item Description	Indirect costs (University of Reading)

Organisation Name	University of Reading

Type of Cost	Other Indirect Costs

Organisation Type	HEI (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£2,531.50	£2,531.50	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£2,532.00	£2,532.00	£5,064.00

Budget Item 5

Item Description	Estates and Technician (Essex)

Organisation Name	University of Essex

Type of Cost	Estate Costs

Organisation Type	HEI (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£2,800.45	£2,800.45	

	Year 1 (£)	**Year 2 (£)**	**Total (£)**
Total rate calculation	£2,800.00	£2,800.00	£5,600.00

Budget Item 6

Item Description	Other indirect costs (Essex)

Organisation Name	University of Essex

Type of Cost	Other Indirect Costs

Organisation Type	HEI (UK)

Rate (%)	100

	Year 1	**Year 2**	
Costs	£17,393.00	£17,393.00	

	Year 1 (£)	Year 2 (£)	Total (£)
Total rate calculation	£17,393.00	£17,393.00	£34,786.00

Justification of costs

This research team brings together the expertise to provide essential insights into the need and supply of the therapy workforce to support children and young people with SEND. The team will be building on existing modelling work and population based approaches within the context of delivering therapy support for CYP within integrated services. Better Communication CIC as the lead organisation is a not-for-profit community interest company with extensive experience of supporting whole systems, commissioning advice and service transformation and provides excellent value for money with no indirect cost charges.

Project team meetings will be held online to reduce travel. Funds are requested to allow two members of the research team to attend a minimum of two national conferences (e.g. British Academy of Childhood Disability) to support dissemination and impact taking account of increased costs to travel and accommodation in the next three years. Costs for catering, travel and subsistence for CYP/families to attend workshops and for researchers - some CYP with more severe disabilities require more than one carer to accompany them for hygiene care, hoisting and overnight accommodation. These costs are based on our experience running workshops for this population. Investigator travel will allow our public co-app and PPI lead to attend workshops.

We also request funds for an encrypted recorder for focus groups and workshops and one laptop for the RA. Funds are requested for workshop materials (i.e., coloured pens, stickers, printing), and accessible material tools adapted for children with disabilities to enable participation.
PPIEP costs cover meetings, refreshments (delivered to homes for online parent meetings), childcare and carer costs, travel (for f2f CYP meetings only), meeting room costs (for f2f CYP meetings only) and payment of parents/carers and CYP time. We have costed for 16 members (8 parents/carers, 2 groups of CYP, one in each region) of to allow for drop out. Costs to cover communication resources to support PPI with parents/carers and CYP with SEND: ‘Talking Mats’ and training to use Talking Mats for PPI lead. Costs cover training and support for PPI group members and PPI member travel costs for dissemination events and PPI-lead/public co-applicant travel costs for CYP PPI groups. Holding parent PPI meetings online will reduce travel and meeting room costs.

We are requesting funds to compensate members of our advisory group for their time to participate in advisory group meetings. Costs are based on standard PPI costs. We are also requesting funds to provide parents and CYP with a thank you voucher for taking part in workshops (120 at @£25 = £3000). Direct costs cover interpreters and voice recording of the PiS to enable inclusion of participants from different communities, accessible room hire. Transcription costs will save researcher time.

Costs are requested to face to face dissemination events, infographics, co-production of a film (with PPI groups) and a website to maximise impact and reach of research findings.

NHS Support costs are needed for informed consent for workshops (WP2) with CYP with SEND and parents/carers recruited via NHS services in both regions

Totals

	Year 1 (£)	Year 2 (£)	Total (£)
Staff Costs	£140,664.09	£143,319.77	£283,983.86
Travel, Subsistence & Conference Costs	£5,026.00	£5,226.00	£10,252.00
Equipment Costs	£1,200.00	£0.00	£1,200.00
Consumables	£1,050.00	£600.00	£1,650.00
PPIEP Costs	£7,249.00	£5,860.00	£13,109.00
Other Direct Costs	£6,682.00	£5,511.00	£12,193.00
Dissemination Costs	£2,840.00	£6,700.00	£9,540.00
Indirect Costs	£83,987.00	£83,987.00	£167,974.00
Total	**£248,698.09**	**£251,203.77**	**£499,901.86**

9. Management & Governance

Is Clinical Trials Authorisation required?	No

Does your project require ethics approval?	Yes

If yes, has ethics approval already been obtained?	No

10. Uploads

Royal College of Speech
and Language Therapists
2 White Hart Yard
London
SE1 1NX

01.05.25

Dear Professor Joffe / Marie Gascoigne,

Subject: NIHR Policy Research Programme (PRP) (38-01-08) - Demand for and supply of therapists for children and young people with special educational needs and disabilities (SEND).

The Royal College of Speech and Language Therapists (RCSLT) is pleased to provide a letter of support for your project looking to understand the demand and supply of therapy for children and young people (CYP) with special educational needs and disabilities (SEND).

The RCSLT is the professional body representing over 20,000 Speech and Language Therapists (SLTs) in the UK and an estimated 2.5 million people who have communication support needs. We facilitate and promote research, promote better education and training of SLTs and provide information for our members and the public about speech and language therapy, speech language and communication needs and swallowing difficulties.

Your proposed project addresses an area currently in crisis: SLT services for CYP with communication and/or swallowing needs. One factor contributing to this issue is our lack of understanding of this area and your project aims to address this issue. As a result, we would be eager to see the results of your proposed research.

If your bid is successful in securing funding, we would be happy to consider supporting dissemination through our communication channels. When ready, we would also be keen to hear about your findings to consider how/whether we can use them in our influencing work with system leaders.

We look forward to hearing the outcome of your funding bid.

Sincerely,

Steve Jamieson, CEO of RCSLT

Irma Donaldson, Chair of RCSLT

Chief Executive: Steve Jamieson MSc BSc (Hons) RN
Patron: Her Royal Highness The Duchess of Edinburgh GCVO, GCStJ, CD
Registered Office: 2 White Hart Yard, London SE1 1NX
Tel: 020 7378 1200 Email: info@rcslt.org

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Incorporated in England No. 518344 • VAT No. 821 1602 77
Registered Charity No. 273724 • Scotland Charity No. SC041191

Royal College of Occupational Therapists

Phoenix House
106-114 Borough High Street
London SE1 1LB
T: 020 3141 4600
rcot.co.uk

National Institute for Health and Care Research
By email.

23/5/25

Letter of Support – Call PRP (37-01-05) Demand for and supply of therapists for children and young people with special educational needs and disabilities (SEND).

Project title: Meeting the needs of children and young people (CYP) (0-18) with Special Educational Needs and/or Disabilities (SEND): modelling the Allied Health Professional (AHP) therapy workforce within integrated children's services.

Primary Investigators: Marie Gascoigne and Victoria Joffe

Dear Colleague

The Royal College of Occupational Therapists has championed the profession and the people behind it for over 80 years; and today, we are thriving with over 36,000 members. Then and now, we're here to help achieve life-changing breakthroughs for our members, for the people they support and for society as a whole. We are excited to see the above multi-disciplinary study proposal, involving occupational therapists, having been shortlisted to Stage 2, and we are very pleased to give our support to the study team. The proposal targets four of our current research priorities for occupational therapy in the United Kingdom and is focused on one of our priority workforce growth areas for the provision of services to children and young people.

If this study is funded, we very much welcome the opportunity to work with the primary investigators and the occupational therapy researchers, Dr Hortensia Gimeno and Sharon Tuppeny, to ensure that the findings and learning from the study will be used within health and care systems across organisations to improve access to SEND services and evaluate effectiveness of therapies for children and young people. We want to see the study results influencing policy and practice and take a leadership role in creating context and drive for change across the system – including NHS professionals, decision makers, service leaders, commissioners, national guideline developers, and working closely with the people who access services and their advocacy organisations so that children and young people with SEND can access the therapy support they need.

We are keen to collaborate with the study team to inform clinical practice and service delivery and

support dissemination of the study and results through our RCOT networks and channels.

With kind regards,

Dr. Gillian Ward
Head of Research and Innovation.

Steve Ford
CEO

Royal College of Occupational Therapists (or 'RCOT') is a registered charity in England and Wales (No. 275119) and in Scotland (No. SC039573) and a limited company registered in England (No. 1347374). Registered office: Phoenix House, 106–114 Borough High Street, London SE1 1LB. VAT Reg. No. 242 7829 47.

Birmingham Community Healthcare
NHS Foundation Trust

Children's and Families Division

BCHC NHS Foundation Trust

Priestly Wharf, Holt Street

Birmingham. B7 4BN

23/05/2025

Re: NIHR Funding Application – Letter of Support

Funding call: *NIHR* Call PRP (37-01-05) Demand for and supply of therapists for children and young people with special educational needs and disabilities (SEND).

Dear NIHR Panel,

I am writing to confirm Birmingham Community Healthcare NHS Foundation Trust's support for the research proposal being led by Better Communication CIC, in partnership with the Universities of Essex, Reading and QMUL, focusing on developing a workforce model for children's therapies.

Our Birmingham system has been supported by Better Communication CIC since 2021 to transform towards a Balanced System® approach for children and young people with SLCN. This has been in the context of very low speech and language therapy staffing ratios relative to the needs of our diverse population.

Despite these challenges we can see positive impact of this approach, we are confident that we are using all our current resources more effectively and are providing an improved Speech and Language Therapy service for the children and families of Birmingham.

It will be so valuable to extend the workforce modelling to include other children's therapies.

We believe this study is well-timed, relevant to the challenges our services face, and has the potential to inform support for children and families in meaningful ways. We hope the application is successful and look forward to playing a role in its delivery.

Yours sincerely,

Hazel Allaway

Professional Lead SLT

Sarah Acton

Group Clinical Manager

Essex County Council

NHS
Hertfordshire and
West Essex
Integrated Care Board

To Whom It May Concern,

We are writing on behalf of Essex County Council and the Essex NHS Integrated Care Boards to express our full support for the research proposal being submitted by Better Communication CIC, in partnership with the University of Essex, Queen Margaret University London, and the University of Reading, in response to the NIHR/DHSC call to develop a model for estimating the workforce required to meet the needs of children and young people (CYP).

This proposal, led by Marie Gascoigne as Principal Investigator and Professor Vicky Joffe as Co-Principal Investigator, brings together a highly skilled and multidisciplinary team. The inclusion of experts such as Dr Hortensia Gimeno, Professor Reinhold Scherer, Dr Emma Pagnamenta, Sharon Tuppeny, and Maxine Darby ensures a comprehensive and innovative approach to this important work. We are also pleased to see the engagement strategy being supported by Jo Harrison and the National Parent Carer Forum (PCF).

We are particularly supportive of the use of the Balanced System® methodology, which is well-aligned with our strategic priorities and ongoing work in Greater Essex. The proposal to include Greater Essex as one of the three modelling sites is welcomed, and we are committed to supporting this research alongside the ongoing transformation program.

This research has the potential to make a significant contribution to national workforce planning and to improving outcomes for children and young people. We are pleased to endorse this proposal and look forward to working collaboratively with the project team.

Yours sincerely,

Kate Martin
SEND Provision Lead
for Contracting & Commissioning
Essex County Council

Rosalind French
Assistant Director of Commissioning for
Children & Young People
Hertfordshire & West Essex ICB

Southend, Essex & Thurrock Children's Therapies Programme

CHARTERED
SOCIETY
OF
PHYSIOTHERAPY

3rd Floor South
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London, EC4A 1AB

Tel **+44 (0)20 7306 6666**
Email **enquiries@csp.org.uk**

www.csp.org.uk

Company registration number:
RC000107

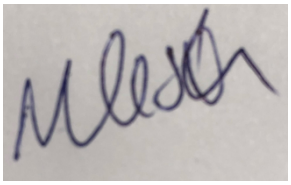
Re: letter of Support for Prof Victoria Joffe

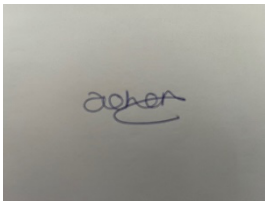
To whom it may concern,

The Chartered Society of Physiotherapy recognise the critical need for robust methods to understand the demand and supply of physiotherapists across the health and social care sector, especially in services for children and young people with special educational needs (SEND). At present, the service requirements and provision for children and young people with SEND is not well understood, leaving many people underserved by limited and fractured services.

We would like to offer our support to Prof. Joffe's application, which we believe will support the development of workforce supply models to ensure that the needs CYP with SEND are met.

Yours Sincerely

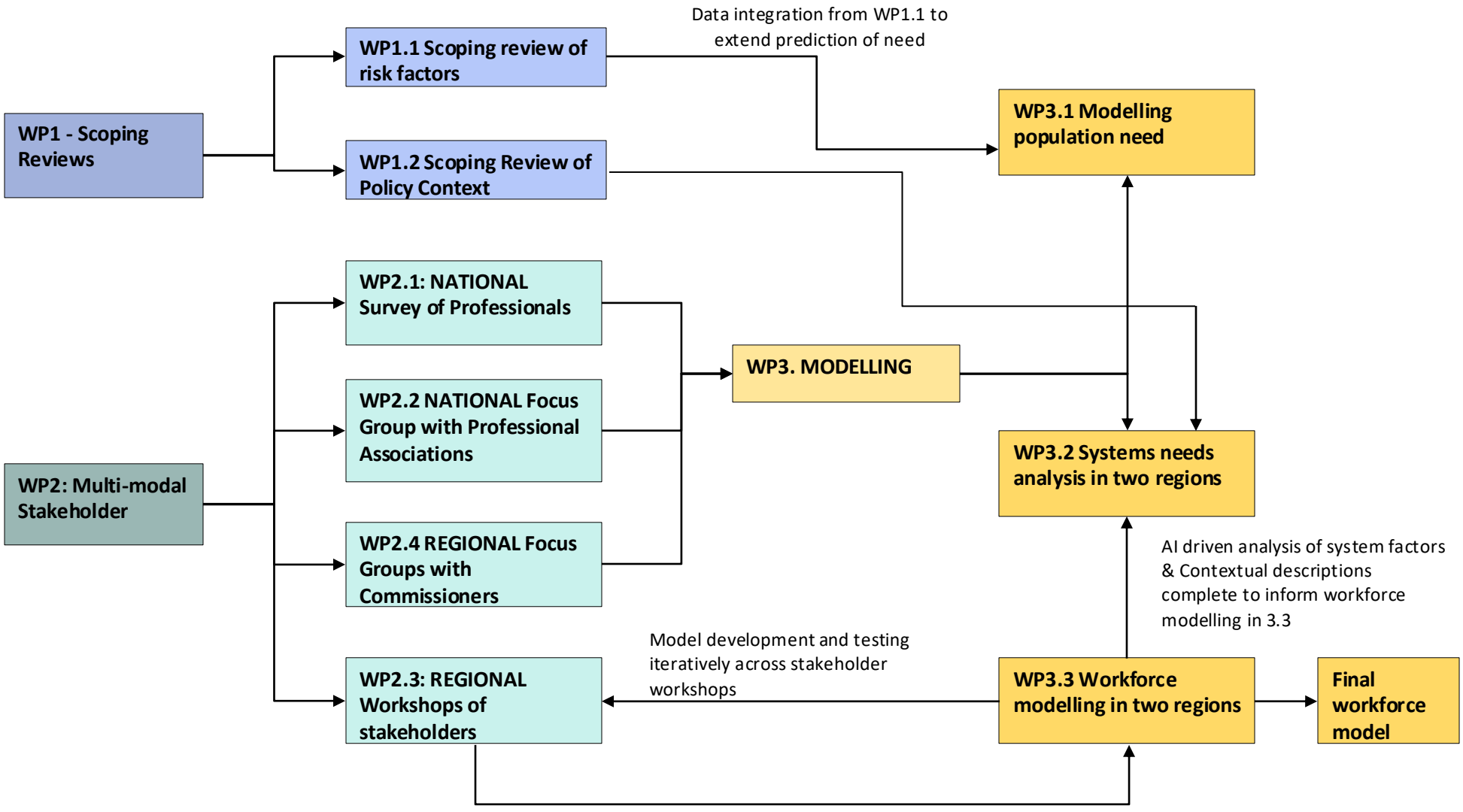




Matthew Liston PhD
Head of Research and Development
Chartered Society of Physiotherapy
listonm@csp.org.uk

Abi Henderson
Head of Workforce and Education
Chartered Society of Physiotherapy
hendersona@csp.org.uk

POSITIVE ABOUT DISABLED PEOPLE


WP1 - Scoping Reviews
WP1.1 Scoping review of risk factors
WP1.2 Scoping Review of Policy Context
Data integration from WP1.1 to extend prediction of need
WP3.1 Modelling population need
WP2: Multi-modal Stakeholder
WP2.1: NATIONAL Survey of Professionals
WP2.2 NATIONAL Focus Group with Professional Associations
WP2.4 REGIONAL Focus Groups with Commissioners
WP2.3: REGIONAL Workshops of stakeholders
WP3. MODELLING
WP3.2 Systems needs analysis in two regions
AI driven analysis of system factors & Contextual descriptions complete to inform workforce modelling in 3.3
Model development and testing iteratively across stakeholder workshops
WP3.3 Workforce modelling in two regions
Final workforce model

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11. Administrative contact details

Administrative contact name	Michele Seidler
Administrative contact job title	Organisational Co-ordinator
Administrative contact telephone number	07752779552
Administrative contact email address	michele.seidler@bettercommunication.org.uk

12. Research & Development office contact

R&D contact name	Marie Gascoigne
R&D contact job title	Director
R&D contact telephone number	07732016568
R&D contact email address	marie.gascoigne@bettercommunication.org.uk

13. Acknowledgement, review and submit

COI declaration Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.
The copyright for the Balanced System® sits with Marie Gascoigne and the research proposal makes use of this System. The Balanced System tools are not in themselves used as a commercial package but are deployed to support commissioned consultancy from Better Communication CIC, a not-for-profit social enterprise.

Lead Applicant - Agreement to terms and conditions

In **ticking** this, you as **Lead Applicant** confirm that the information given on this form is complete and correct and that you will be actively engaged in this research and responsible for its overall management. In addition, you will accept responsibility for ensuring that the host institution and interested parties are kept informed. **Ticking this box constitutes an electronic signature of the Lead Applicant with regard to this application**
☒Confirmed

Checklist of information to include when submitting a NIHR Stage 2 research application
Applicants should click the checkboxes to indicate that they have included the necessary information prior to submitting their application.

Appropriate and relevant involvement of patients and the public
☒Confirmed

A good quality Plain English Summary	☒Confirmed

A clear description of team member roles and contribution
☒Confirmed

A clear scientific abstract
☒Confirmed

A clear description of the changes from first stage
☒Confirmed

A flow diagram illustrating the study design / flow of participants (document upload), if appropriate
☒Confirmed

A full and accurate detailed budget breakdown
☒Confirmed

A clear justification of costs / value for money
☒Confirmed

References (document upload)
☒Confirmed

A clear Detailed Research Plan outlining the study design, methods, dissemination etc.
☒Confirmed

A CTU letter of support if required (document upload)
☒Confirmed

Completed Schedule of Events Cost Attribution Tool (SoECAT)
☒Confirmed

14. CV - Lead Applicant(s)

Full Name	Ms Marie Gascoigne	**Position**	Director
Department		**Institution**	BETTER COMMUNICATION CIC
ORCID iD	0009-0002-2755-7725	**Telephone No.**	+44 7732016568
Address Line 1	Lower Judfalls	**Country**	United Kingdom (England)
Address Line 2	Knowles Brow	**Postcode**	BB7 9PR
Address Line 3		**Web Page.**	www.bettercommunication.org.uk

Qualification	Subject	Institution	From	To
MBA	Masters in Business	Open University (The)	September	June

	Administration		2000	2003
MSc	Human Communication	City, University of London	October 1990	June 1993
BSc	Speech and Language Therapy	The Royal Central School of Speech and Drama	September 1985	June 1989

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

Selected recent relevant publication(s)

Publications
Marie Gascoigne. (2024) 'Meeting speech, language and communication needs: a whole-systems, population-based approach', *Paediatrics and Child Health*.
(2022) 'Equity for All: Children's Speech and Language Services in Scotland: Report for Scottish Government', *Scottish Government*.
James Law, Jenna Charlton, Julie Dockrell, Marie Gascoigne, Cristina McKean and Anna Theakston. (2017) 'Early Language Development: Needs, provision, and intervention for preschool children from socio-economically disadvantage backgrounds A Report for the Education Endowment Foundation', *Education Endowment Foundation* .
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Marie Gascoigne. 'A whole systems approach to meeting speech, language and communication needs', *McCool, S. (Ed) (in press) Supporting Children's Speech, Language and Communication Needs. London: Routledge.*.
Marie Gascoigne. 'So what? And Prove It! Evidencing impact for children and young people', *McCool, S. (Ed) (in press) Supporting Children's Speech, Language and Communication Needs. London: Routledge.*.

Joint Lead Applicant

Full Name	Professor Victoria Joffe	**Institution**	University of Essex
Web Page.	https://www.essex.ac.uk/people/joffe12809/victoria-joffe		

Education

From	To	Qualification	Subject	Country	Organisation	Department
09/1992	06/1998	Dphil	Experimental Psychology	United Kingdom (England)	University of Oxford	Department of Experimental Psychology
01/1987	12/1990	BA(Hons)	Speech and Hearing Therapy		University of the Witwatersrand	Speech Therapy and Audiology

Publications

(2019) 'Joffe, V.L., Rixon, L., and Hulme, C. (2019). Improving storytelling and vocabulary in secondary school students with language disorder: a randomised controlled trial. International Journal of Language and Communication Disorders.',
Murphy, S., Joffe, V., Donald, L., Radley, J., Suntharajah, S., Welch, C., Bell, K., Messer, D., Crafter, S., Fairhurst, C., Corbacho, B., Rodgers, S. and Torgerson, D.. (2021) 'Evaluating 'Enhancing Pragmatic Language skills for Young children with Social communication impairments' (E-PLAYS): a feasibility cluster-randomised controlled trial.', *Pilot and feasibility studies*, 7(1).
Stojanovik, V., Pagnamenta, E., Samson, S., Sutton, R., Jones, B., Joffe, V., Harvey, K., Pizzo, E. and Rae, S.. (2023) 'Evaluating 'An Early Social Communication Intervention for Young Children with Down Syndrome' (ASCEND): Results from a Feasibility Randomised Control Trial',
Pagnamenta, E., Longhurst, L., Breaks, A., Chadd, K., Kulkarni, A., Bryant, V., Tier, K., Rogers, V., Bangera, S.,

Publications

Wallinger, J., Leslie, P., Palmer, R. and Joffe, V.. (2022) 'Research priorities to improve the health of children and adults with dysphagia: a National Institute of Health Research and Royal College of Speech and Language Therapists research priority setting partnership.', *BMJ Open*, 12(1).
Millard, SK., Murphy, S., Barton, G., Rixon, L., Shepstone, L., Sims, E. and Joffe, V.. (2025) 'Palin Stuttering Therapy for School aged Children and usual treatment: A randomised controlled trial feasibility study.', *Journal of fluency disorders*, 84.
Victoria Joffe. (2024) 'An Early Social Communication Intervention for Young Children with Down Syndrome' (ASCEND): Results from a Feasibility Randomised Control Trial. ', *Pilot Feasibility Studies* .

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/04/2024	18	153,712.00	CCF	Harnessing portable smart-camera technology (p-SCT) to support the communication skills of people with aphasia	Co Applicant
01/03/2022	36	518,000.00	East Suffolk and North Essex Foundation Trust.	Fundamentals of care in the acute sector/community: Exploration of "Time to care", and mental health of the physically unwell	Primary Investigator
01/09/2021	48	1,461,792.00	PHR NIHR	EPLAYS trial; evaluation of a computerised intervention to promote communicative development and collaborative skills in children.	Co Investigator
01/08/2019	32	271,013.00	CCF	Evaluating an early social communication intervention for young children with Down syndrome: a feasibility study	Co Applicant
01/06/2019	39	321,121.00	CCF	Evaluating Palin Stammering Therapy for Children: a feasibility study	Co Applicant
01/11/2017	21	252,738.00	CCF	Evaluating 'Enhancing Pragmatic LAnguage skills for Young children with Social communication impairment' (E-PLAYS): A feasibility study	Co Applicant

15. CV - Co-applicants

Co Applicant - 1

Full Name Miss Maxine Darby

Organisation BETTER COMMUNICATION CIC

Education

From	To	Qualification	Subject	Country	Organisation	Department
10/2022	12/2025	MSc	Public Health	United	Brunel	Health

Education

From	To	Qualification	Subject	Country	Organisation	Department
				Kingdom	University	Sciences
10/1996	06/1999	BSc	physiotherapy	United Kingdom	Brunel University	Health Sciences

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

Co Applicant - 2

Full Name Dr Hortensia Gimeno **Organisation** Queen Mary University of London

Education

From	To	Qualification	Subject	Country	Organisation	Department
03/2017	07/2019	PhD	Paediatric Neuro-Rehabilitation	United Kingdom	King's College London	Institute of Psychology, Psychiatry and Neurosciences
09/2002	06/2007	MSc	MSc OT (Paeds)	United Kingdom	University of East London	Occupational Therapy
09/1994	07/1997	BSc	Occupational Therapy	Spain	Universidad Ciencias de la salud, Zaragoza. Spain	Health Sciences

Publications

Gimeno, H. and Polatajko, H.. (2025) 'The Cognitive Orientation to daily Occupational Performance approach in childhood-onset disabilities.', *Developmental medicine and child neurology*.
Lumsden, DE., Tsagkaris, S., Cleary, J., Champion, M., Mundy, H., Mostofi, A., Hasegawa, H., McClelland, VM., Bhattacharjee, S., Silverdale, M., Gimeno, H., Ashkan, K., Selway, R., Kaminska, M., Hammers, A. and Lin, JP.. (2025) 'Outcomes of deep brain stimulation surgery in the management of dystonia in glutaric aciduria type 1.', *Journal of neurology*, 272(3).
Cioffi, E., Hutber, A., Molloy, R., Murden, S., Yurkewich, A., Kirton, A., Lin, JP., Gimeno, H. and McClelland, VM.. (2024) 'EEG-based sensorimotor neurofeedback for motor neurorehabilitation in children and adults: A scoping review.', *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*, 167.
Fehlings, D., Agnew, B., Gimeno, H., Harvey, A., Himmelmann, K., Lin, JP., Mink, JW., Monbaliu, E., Rice, J., Bohn, E. and Falck-Ytter, Y.. (2024) 'Pharmacological and neurosurgical management of cerebral palsy and dystonia: Clinical practice guideline update.', *Developmental medicine and child neurology*, 66(9).
Bates, L., Taylor, M., Lin, JP., Gimeno, H., Kingston, J. and Rudebeck, SR.. (2021) 'Mental health and behaviour in children with dystonia: Anxiety, challenging behaviour and the relationship to pain and self-esteem.', *European journal of paediatric neurology : EJPN : official journal of the European Paediatric Neurology Society*, 35.
Gimeno, H., Polatajko, HJ., Lin, JP., Cornelius, V. and Brown, RG.. (2021) 'Cognitive Strategy Training in Childhood-Onset Movement Disorders: Replication Across Therapists.', *Frontiers in pediatrics*, 8.

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/09/2024	52	1,446,095.00	National Institute for Health and Care Research	Addressing critical gaps in childhood-onset hyperkinetic movement disorders: Towards evidence-based interventions and	Lead (NIHR Advanced Fellowship)

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
				personalised rehabilitation.	
01/05/2024	18	74,942.00	Barts Charity	Priority outcomes for patients with childhood-onset dystonia in East London and clinicians	Lead applicant
01/10/2023	24	238,261.00	Barts Charity	Improving access to healthcare: Face-to-face And Telehealth Equivalence of assessments in Aphasia (FATE-A)	Collaborator
01/09/2023	36	3,068,526.00	National Institute for Health and Care Research	NIHR INSIGHT: Inspiring Students into Research (R1 2023) (North London)	Co-applicant
01/07/2023	12	16,835.00	Royal College of Occupational Therapiss	Paediatric occupational therapy in the UK: A survey of usual practice.	Co-lead
01/07/2022	36	199,862.00	Rosetrees Trust	Enhancing Cortical Sensoriomotor Processing in Children with Dystonic Cerebral Palsy	Co-applicant
01/07/2021	24	94,186.00	Elizabeth Casson Trust	Can a targeted knowledge translation intervention increase implementation and adoption of evidence in practice	Co-applicant
01/09/2020	9	19,979.00	Health Education England	HEALTH EDUCATION ENGLAND Post-doctoral Bridging Award (personal award)	Lead applicant
01/10/2019	36	3,998,353.00	European Council Horizon	Personalised recovery through a multi-user environment: virtual reality for rehabilitation	Co-applicant
01/03/2014	36	313,047.00	National Institute for Health and Care Research	Cognitive Orientation to daily Occupational Performance Intervention with Paediatric Dystonic Movement Disorders (NIHR Doctoral Fellowship)	Lead

Co Applicant - 3

Full Name Miss Sophia TA Hill **Organisation** PPI representative

Education

From	To	Qualification	Subject	Country	Organisation	Department

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant

Co Applicant - 4

Full Name Dr Emma Pagnamenta **Organisation** University of Reading

Education

From	To	Qualification	Subject	Country	Organisation	Department
08/2004	09/2006	MSc	Speech and Language Sciences	United Kingdom	University College London	Psychology and Language Sciences
10/2000	02/2004	PhD	Neuroscience	United Kingdom	University College London	Developmental Cognitive Neuroscience, Institute of Child Health
10/1999	10/2000	MSc	Neuroscience	United Kingdom (Scotland)	University of Edinburgh	Neuroscience
10/1996	07/1999	BSc	Psychology	United Kingdom	University of Durham	Psychology

Publications

Naomi Leafe, Emma Pagnamenta, Mark Donnelly, Laurence Taggart and Jill Titterington. (2025) ''To Know That You Are a Link in the Chain': A Realist Evaluation to Explore How Digital, Intensive, Parent-Implemented Interventions Work for Children With Speech Sound Disorder, Why, and for Whom', *International journal of language & communication disorders*.
Leafe, N, Emma Pagnamenta, Taggart, L, Donnelly, M, Hassiotis, A and Titterington, J. (2025) 'What works, how and in which contexts when using digital health to support parents/carers to implement intensive speech and language therapy at home for children with speech sound disorder? A realist review', *PLoS ONE*.
Katherine Pritchard, Vesna Stojanovik, Jill Titterington and Emma Pagnamenta. (2025) 'How speech and language therapists and parents work together in the therapeutic process for children with speech sound disorder: A scoping review', *International journal of language & communication disorders*.
Vesna Stojanovik, Emma Pagnamenta, Sampson, Sarah, Sutton, Rachel, Jones, Benjamin, Joffe, Victoria, Kate Harvey, Pizzo, Elena and Rae, Sarah. (2024) 'Evaluating an early social communication intervention for young children with Down Syndrome (ASCEND): results from a feasibility randomised control trial', *Pilot and feasibility studies*.
Gino Hipolito, Emma Pagnamenta, Helen Stacey, Emily Wright, Victoria Joffe, Kou Murayama and Cathy Creswell. (2023) 'A systematic review and meta-analysis of nonpharmacological interventions for children and adolescents with selective mutism', *JCPP Advances*.
Emma Pagnamenta, Lauren Longhurst, Anne Breaks, Katie Chadd, Amit Kulkarni, Val Bryant, Kathy Tier, Vanessa Rogers, Sai Bangera, Josephine Wallinger, Paula Leslie, Rebecca Palmer and Victoria Joffe. (2022) 'Research priorities to improve the health of children and adults with dysphagia: a National Institute of Health Research and Royal College of Speech and Language Therapists research priority setting partnership', *BMJ Open*.

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/08/2021	24	425,115.00	ESRC	Parent-Delivered Early Language Intervention for Children with Down Syndrome: A Randomised Controlled Feasibility Trial	Co-applicant
01/08/2019	32	271,013.00	CCF	Evaluating an early social communication intervention for young children with Down syndrome: a feasibility	Co Applicant

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
				study	

Co Applicant - 5

Full Name Professor Reinhold Scherer **Organisation** University of Essex

Education

From	To	Qualification	Subject	Country	Organisation	Department
05/2010	01/2016	Habilitation (Venia Docendi)	Applied Computer Science	Austria	Technische Universität Graz	Graz University of Technology
09/2001	03/2008	PhD	Computer Science	Austria	Technische Universität Graz	Graz University of Technology
10/1991	06/2001	MSc(Hons)	Computer Science	Austria	Technische Universität Graz	Graz University of Technology

Publications

Reinhold Scherer, Mohammadreza Jamalifard, Benjamin R. Underwood, Jonathan R. Lewis, Alison Skea, Elspeth Mathie, Karen Harrison Dening, Steve Milton, Claire Surr, Reda M. Lebcir, Amit N. Pujari, Robert C. Free, Theng Yin Leng, Michael Craven, Lucy Beishon, Victoria L. Keevil, Claire Goodman and Melanie Handley. 'Understanding dynamics to Streamline Access: Systems Engineering and Agent-Based Probabilistic Modelling in Memory Assessment Services',
Noemi Massetti, Mirella Russo, Raffaella Franciotti, Davide Nardini, Giorgio Maria Mandolini, Alberto Granzotto, Manuela Bomba, Stefano Delli Pizzi, Alessandra Mosca, Reinhold Scherer, Marco Onofrj, Stefano L. Sensi, for the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Alzheimer's Disease Metabolomics Consortium (ADMC). (2022) 'A Machine Learning-Based Holistic Approach to Predict the Clinical Course of Patients within the Alzheimer's Disease Spectrum1', *Journal of Alzheimer's Disease*.
Reinhold Scherer. (2019) 'Thought-based interaction: Same data, same methods, different results?', *PLoS biology*.
Lan, Z., Sourina, O., Wang, L., Scherer, R. and Muller-Putz, G.R.. (2019) 'Domain Adaptation Techniques for EEG-Based Emotion Recognition: A Comparative Study on Two Public Datasets', *IEEE Transactions on Cognitive and Developmental Systems*.
Martin Seeber, Reinhold Scherer and Gernot R. Müller-Putz. (2016) 'EEG Oscillations Are Modulated in Different Behavior-Related Networks during Rhythmic Finger Movements', *The Journal of Neuroscience*.
Chavarriaga, R., Fried-Oken, M., Kleih, S., Lotte, F. and Scherer, R.. (2016) 'Heading for new shores! Overcoming pitfalls in BCI design.', *Brain computer interfaces (Abingdon, England)*, 4(1-2).

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
01/04/2024	18	153,712.00	CCF	Harnessing portable smart-camera technology (p-SCT) to support the communication skills of people with aphasia	Co Applicant
01/07/2023	18	202,778.00	National Institute for Health Research (NIHR)	MEDAL - Innovation Hub for agile development and optimization of health and care services for people with multiple long-term conditions including dementia and their caregivers	Joint Lead Applicant
01/09/2021	9	11,114.00	Union of European Football Associations (Basel, CH)	Exploring the Decision-Making of Video Assistant Referees (VAR)	Co-I

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant
				in Football	
01/01/2021	24	126,000.00	Azienda Sanitaria Dell'Alto Adige (Vipiteno, IT)	Study of Functional Electrical Stimulation with assistive support driven by a Brain-Computer Interface on the upper limb rehabilitation of chronic stroke patients	Co-I

Co Applicant - 6

Full Name Mrs Sharon Tuppeny **Organisation** BETTER COMMUNICATION CIC

Education

From	To	Qualification	Subject	Country	Organisation	Department
10/2021	05/2025	MSc	Leading Quality Improvement	United Kingdom	University of Bangor	Health Sciences
10/1991	06/1993	Post Graduate Diploma	Occupational Therapy	United Kingdom	Queen Mary University of London	Medical school
10/1986	06/1989	BA(Hons)	Economics and economic history	United Kingdom	University of Sussex	school of social sciences

Publications

Grants

Start Date	Duration	Amount	Source	Title	Role of Applicant